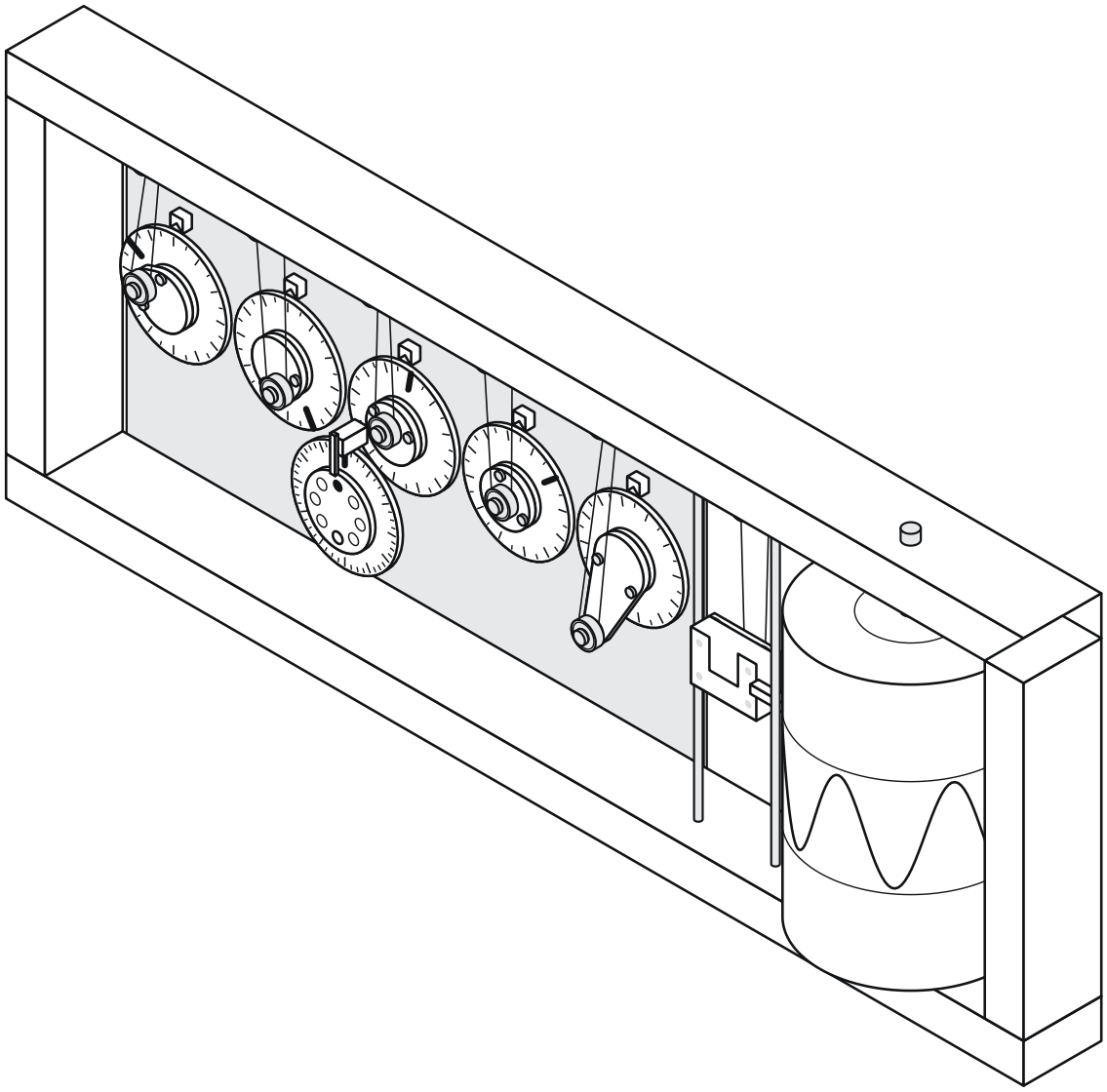
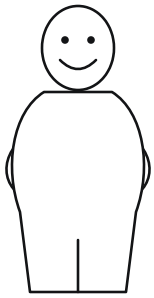
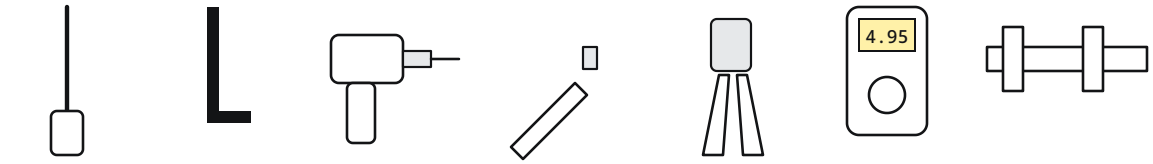
# Tide Machine

**A five-constituent harmonic tide predictor: five stepper-driven arms sum on one line and draw the tide on a drum. Assembly, board, smoke test, wiring, first power.**

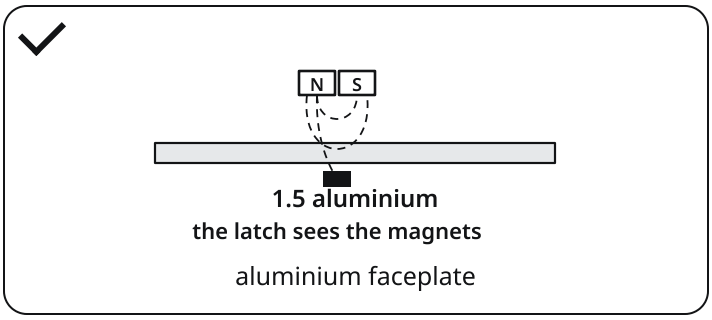


★ **Another Amazing DIY Project**
by Mike Montana

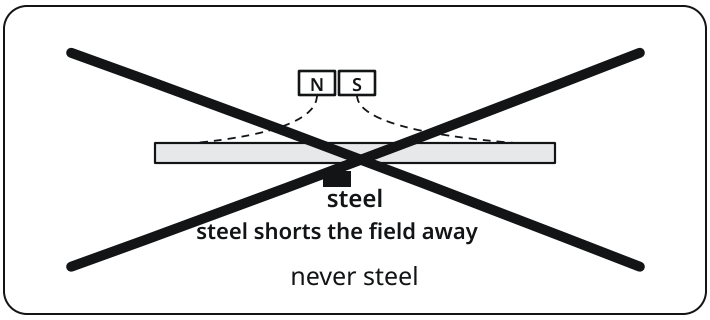




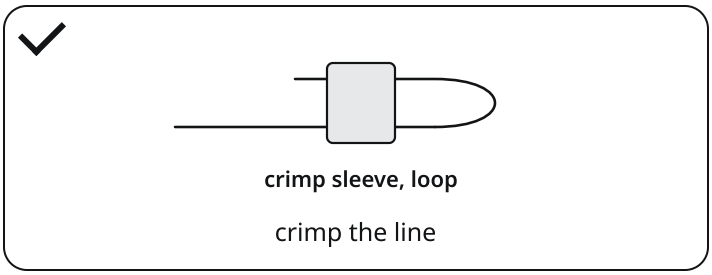

PH1 driver · hex 2, 2.5 · drill, 3.1 bit · iron + insert tip · XH crimper · multimeter · calipers



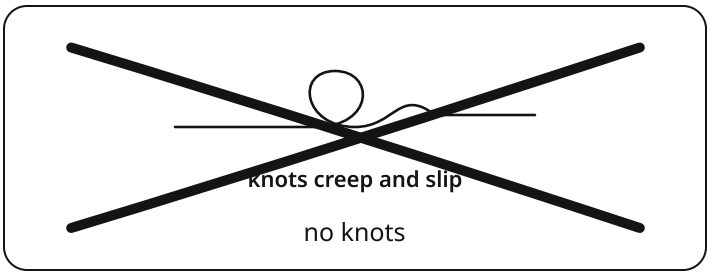

aluminium faceplate




never steel




crimp the line




no knots

**Datum.** Every number is in the machine frame: **x** along the row (S2 at −224 … M2 at 0, pen +70, drum +155.5), **y** up, **z** negative toward you. The faceplate's front face is z = 0. The figures are drawn from `station_study.py`, the same model the fit checks run on.

**Never turn an arm at constant speed**, not even to test it by hand with a drill: the firmware drives each arm at an uneven rate so the line length is exactly sinusoidal.

## Contents


**A** **Mechanics**
steps 1 to 16

**B** **Main board**
steps 17 to 33

**C** **Smoke test**
steps 34 to 40

**D** **Wiring**
steps 41 to 47

**E** **First power-up, calibration**
steps 48 to 56

**F** **How to use**
steps 57 to 60

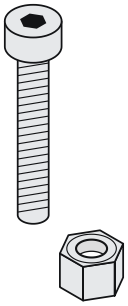
**G** **Appendix: schematics**
system, logical, full board

**!** **Known issues**
read before you order or cut


Give feedback by **step number**; they won't change. Status: the mechanics are Rev M (fit-checked, not yet built), the board is rev B (routed, DRC clean, not yet ordered), and the firmware isn't written, so section E is mostly **planned**.

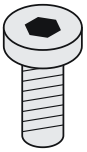
# 1

## Fasteners



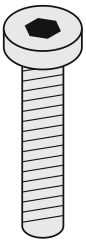
**6x**
pulley-bracket axles

M3 × 20 + nyloc



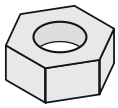
**5x**
arm-bearing axles

ISO 7380 M3 × 8



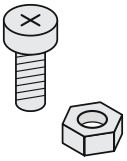
**5x**
pen pulley + 4 carriage axles

ISO 7380 M3 × 16



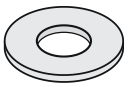
**4x**
carriage axles, rear face

ISO 4032 M3



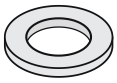
**12x**
motor ears; head 2.4 tall or less

M3 × 8 pan + nut



**12x**
under the ear nuts only

M3 washer



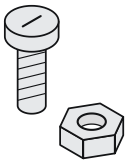
**16x**
brackets 6, carriage 4, arm axles 5, pen pulley 1

3 × 5 × 0.5 shim



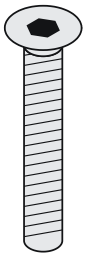
**6x**
5 arms + carriage front

M3 × 4 heat-set



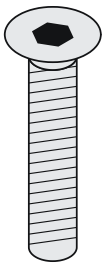
**30x**
arms, Hall and calendar carriers, tabs, hand

M2 × 6 + nut



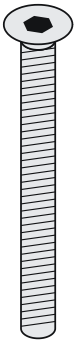
**12x**
pulley brackets

#4 × 3/4″ flat



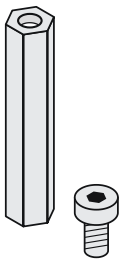
**10x**
faceplate to the rails

#6 × 3/4″



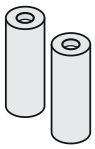
**4x**
end posts ( KI 9 )

#6 × 1-1/2″



**4x**
main board standoffs, 8 screws

M3 × 30 F-F + M3 × 6



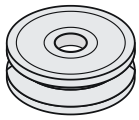
**4x**
line ends

crimp 0.6–0.8 ID

**Shims, not washers, against a bearing.** An M3 washer is Ø7 and presses on the V623ZZ's shield; the inner race is only Ø5. Every bearing face that gets clamped gets a 3 × 5 × 0.5 shim. That's 16, one more than the 15 the BOM lists ( KI 6 ).

# 2

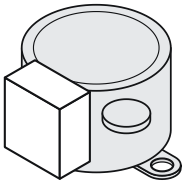
## Motion, sensing, frame



V623ZZ 3 × 12 × 4

**22x**

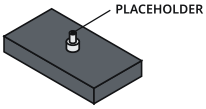
12 rail, 5 arms, pen pulley, 4 carriage



28BYJ-48 5 V

**6x**

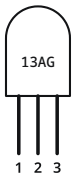
five stations + the drum




BKA30D-R5

**1x**

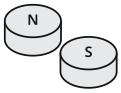
calendar, dual concentric shafts ( **KI 5** )




DRV5013 TO-92

**7x**

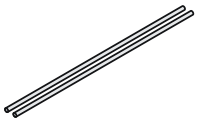
Hall latches: 5 stations, year, moon




N52 5 × 2 axial

**14x**

seven opposed pairs



3 mm h6 × 170

**2x**

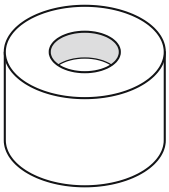
pen guide rods



0.30 fluorocarbon

**2 m**

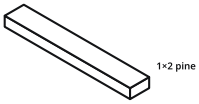
the line; not nylon



3″ Sch 40 PVC

**130**

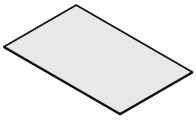
the drum, cut square; 7 mm shaft




1 × 2 pine

**1302**

rails 2 × 499, posts 2 × 152



5052 Al 1.5

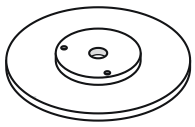
**1x**

faceplate 283 × 176, laser cut ( **KI 1** )

Also: perfboard (70 × 90, double-sided) for the carriers, 3-core 26 AWG and 8-core lead, JST XHP housings and crimps, PLA, clear silicone (fallback only), decal vinyl, a 0.3–0.5 mm fibre-tip pen.

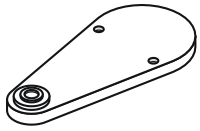
# 3

## Printed parts, as they sit on the bed



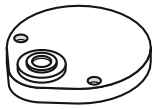
hub_dial.stl

**5x**
dial back down, boss up



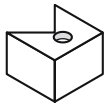
arm_M2.stl

**1x**
r 35.00; back down



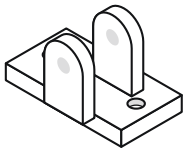
arm_S2 / N2 / K1 / O1.stl

**4x**
r 8.75, 7.35, 3.50, 2.45



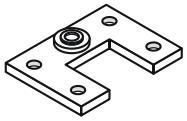
**5x**
flat

index_tab.stl



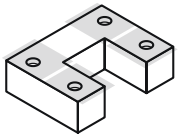
pulley_bracket.stl

**6x**
rail face down



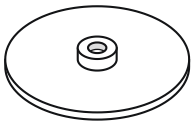
pen_carriage_rear.stl

**1x**
pulley stub up



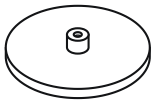
pen_carriage_front.stl

**1x**
pockets open upward



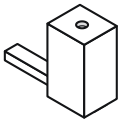
year_dial.stl

**1x**
decal face down ( **KI 5** )



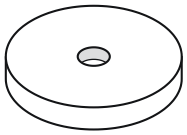
moon_disc.stl

**1x**
decal face down ( **KI 5** )



calendar_hand.stl

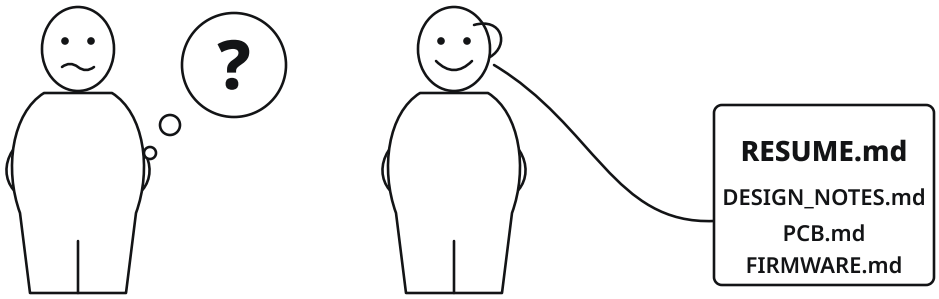
**1x**
front face down



drum_end_cap.stl

**2x**
flat ( **KI 8** )

**Print the coupons first**: the D-bore (tune it to a light press on a motor's flats), a magnet pocket at 5.1 and 5.2, the 4.0 insert hole, and a PVC-bore ring for the drum cap. PLA, 3 perimeters, no supports anywhere. The files are in `stl/`, regenerated by `export_stl.py` from the same model as these drawings.




## Known issues found while writing this guide

**check before you order or cut**

**KI 1** **No Rev M cut files yet.** `tide_machine.rb`, `faceplate.py` and `rails.py` still describe Rev L. The hole schedule in step 1 comes from `station_study.py`. Port it to the Ruby before ordering the plate or drilling the rails. The #6 frame-screw holes (Ø4.0 at y ±82) have no spacing yet. Step 1, step 14.

**KI 2** **Hall windows are still sized for the SOT-23.** They're 3.4 × 3.4; the TO-92 latch needs about 4.4 × 3.4, and its sensing element sits a few tenths deeper. Re-run the field check (the magnet gap was 4.40 mm) when you resize them. Step 1, step 5.

**KI 3** **No holes for the main board yet.** Proposed: 4 × Ø3.2 at (25, 39), (−108, 39), (25, −35.28), (−108, −35.28), for the board upright as seen from behind. Checked against the whole model (the standoffs, the board and the lead ends under it): nothing within 3 mm. Screw heads under a dial are fine at 2.4 mm tall or less (the same 1.1 mm as the motor-ear screws). Step 41.

**KI 4** **The Hall carriers and the calendar carrier are hand-cut perfboard**, not designed parts. As a plain 36 × 68 rectangle, the calendar carrier's top corners clipped the N2 and K1 motor bodies (8.9 $mm^3$ each; each corner is 11.7 mm from a motor's axis, against its 14 mm radius). **Fixed in the model:** cut 4 × 6 off both top corners; it now clears by 0.98 mm. **Still open:** the two bottom holes at y −78 are under the bottom rail (its inner face is at y −76), so their M2 heads have nowhere to go on the front; and all four nuts sit inside the BKA placeholder's outline. Move the holes once the motor is measured. Step 5, step 6.

**KI 5** **BKA30D-R5 is a placeholder.** The body, the shaft diameters, lengths and positions, and whether both shafts turn continuously (the family datasheet shows stops) are all unconfirmed. The year dial's and moon disc's hub bores are placeholders, and so is the coil-to-pin map in step 45. Step 6, step 10, step 45.

**KI 6** **Arm-bearing axles clamped a washer on the shield.** The M3 × 8 went through an M3 washer (Ø7) against the V623ZZ's face; the inner race is Ø5, so 1 mm of the washer bore on the shield. The pen pulley's M3 × 16 head (Ø5.7) did the same by 0.35 mm. **Fixed in the model:** a 3 × 5 × 0.5 shim under each (16 shims in all). The pen pulley's screw now engages its insert by 2.25 mm. Step 9, step 12.

**KI 7** **The pen arm ran through the +x guide rod** (27.9 $mm^3$) at the drum's mid-plane (z −22 to −26, rod at −23.75). **Fixed in the model:** the arm runs 1 mm in front of the rods, at z −26.25 to −30.25; the nib meets the drum at x 111.2. The pen holder, leaf spring, retract lever and ballast pocket still aren't designed. Step 12.

**KI 8** **Drum drive not designed.** Nothing is drawn yet for the motor mount, the coupling to the 7 mm shaft, or how the end caps grip the shaft (the cap STL only has a Ø7.1 bore). The drum motor sits

outside the top rail, about 18 cm in a straight line from J10; its lead is about 24 cm. Route it tight or make a 5-way XH extension. Step 15, step 43.

**KI 9** **End-post screws are too short.** The BOM's #6 × 3/4″ can't reach through a 19 mm rail into the post's end grain. Use #6 × 1-1/2″ (4) with glue; the faceplate still takes #6 × 3/4″ (10). Step 13.

**KI 10** **The far dead end has no fixing.** Nothing slides at S2's IN pulley, so the line can end there, but no tie-off point is designed. The pen bracket's screw-adjustable anchor, copied onto S2's bracket, would do. Step 16.

**KI 11** **Decals use placeholder constituents.** Put NOAA G values, the longitude and the epoch into `decals.py`, and check the K1/O1 ±90° convention (it moves those zeros by 12 h), before printing. Step 8.

**KI 12** **No firmware yet.** Everything in section E and F tagged **planned** needs it: the inverse tables, the Hall edge timing, homing, NTP, the PWM hold. Section E.

**KI 13** **No 1:1 rail templates yet.** `rails.py` printed drilling templates for Rev L; Rev M needs new ones from the model, printed at 100% and tiled across Letter pages (the rail is 498.9 long), with a calibration bar, page-join marks, and each sheet labelled top or bottom rail, face, and S2 end. Until then, measure from the **S2 end**. Top rail, inner face: #4 pilots at 50, 106, 162, 218, 274 and 344 (each ± 11), 15 from the rear edge; rod sockets Ø3.1 × 9 at 325.08 and 362.92, 23.75 from the rear edge (the bottom rail's inner face takes the same two). Both rails: the drum hole at 429.45, 24 from the rear edge. Rear faces: faceplate pilots 6 from the inner edge (spacing with KI 1). Outer faces: end-post screws 9.5 from each end. Step 11, step 13.

## Fixed in the model while writing this guide

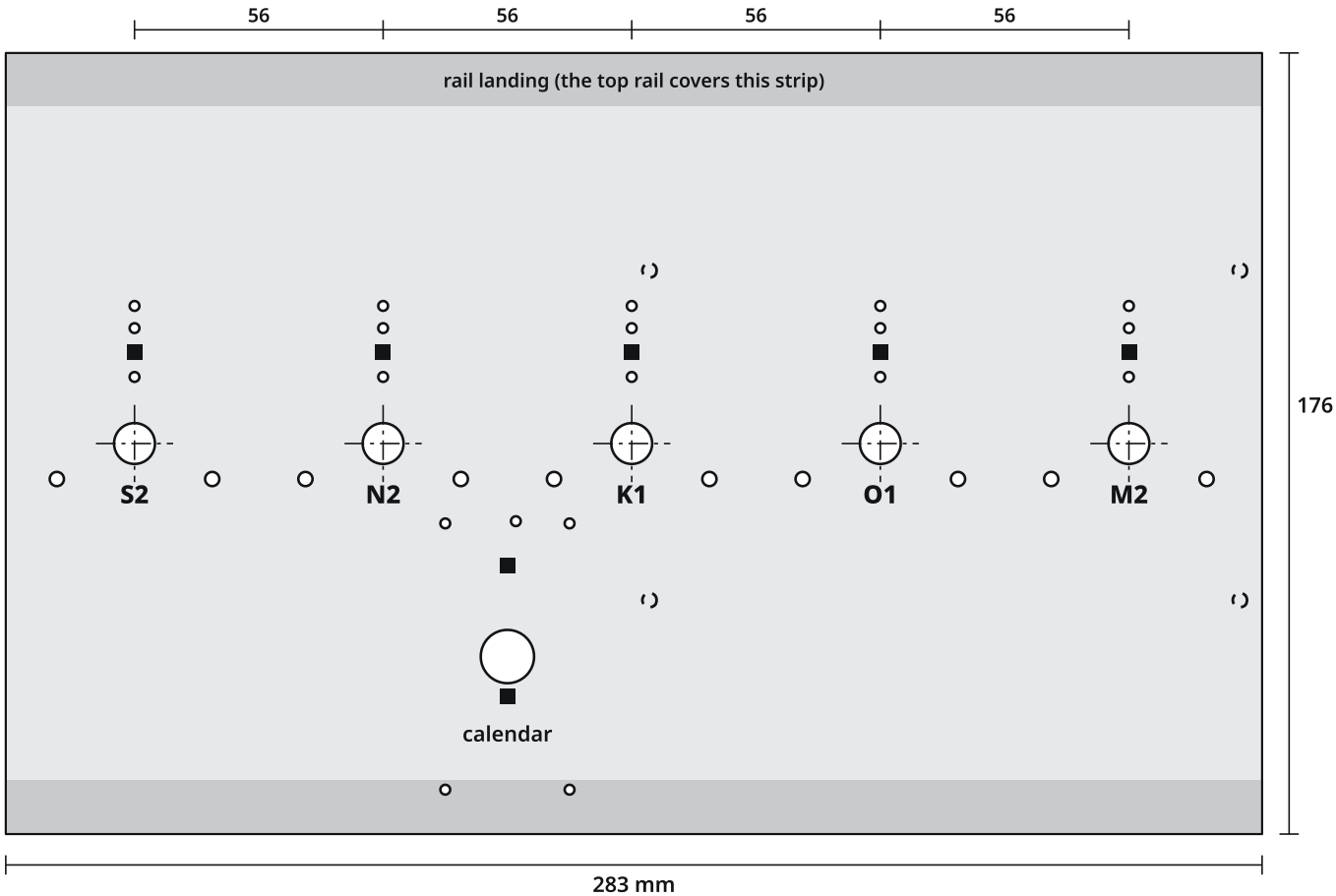
- The calendar sat at x −196 in the fit study but −140 in the spec; now −140 everywhere.
- The left end post ran into S2's dial, magnets and ear screws; the rails grew 12 mm (now 499; left post 5 mm clear of the dial).
- The calendar index hand's base overlapped N2's dial; it was narrowed and moved down (1.09 mm clear).
- The index tab and the calendar hand had no pilot holes for their M2 screws; both STLs now have a Ø1.6 pilot.
- KI 4, KI 6 and KI 7 above. Each fix was confirmed with a real interference check (manifold intersections of every pair of parts, `check_interference.py`) and the fit study's 31 clearances, all ≥ 0.98 mm.

# A Mechanics

Plate work first: behind it (motors, carriers), then in front (dials, arms). Then the frame, with the pen carriage inside it, and the plate onto the frame. The line goes on last.

## 1 Check the faceplate

KI 1 KI 2 KI 3




seen from the FRONT. dashed circles: proposed main-board standoff holes (KI 3)

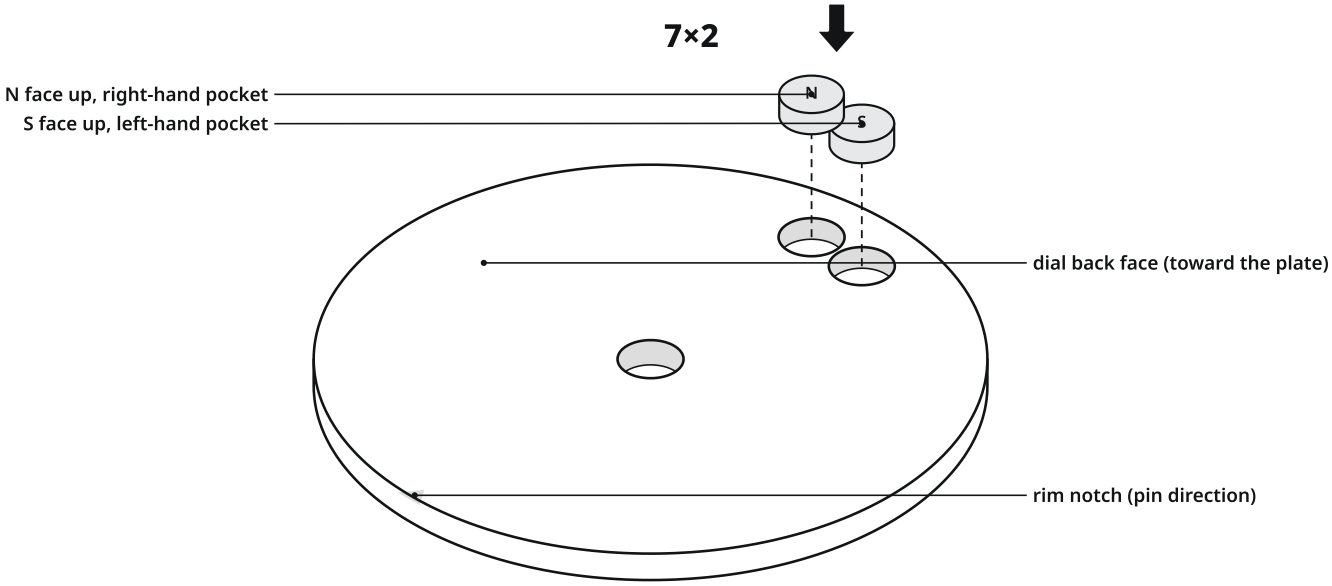
| Count | Size | Where (per station at shaft x, 0) | For |
|---|---|---|---|
| 5 | Ø9.2 | (x, 0) | locates each motor’s Ø9 boss |
| 10 | Ø3.2 | (x ± 17.5, −8) | motor ears, M3 |
| 5 | window 3.4 ( **KI 2** ) | centred (x, 20.5) | the Hall latch nests in it |
| 10 | Ø2.2 | (x, 15) and (x, 26) | Hall carrier, M2 |
| 5 | Ø2.2 | (x, 31) | index tab, M2 |
| 1 | Ø12 | (−140, −48) | calendar shafts and hub adapters |
| 2 | window 3.4 | (−140, −27.5), (−140, −57) | year and moon latches |
| 4 | Ø2.2 | (−140 ± 14, −48 ± 30) | calendar carrier |
| 1 | Ø2.2 | (−138.1, −17.5) | calendar index hand |
| 4 | Ø3.2, proposed | (25, 39), (−108, 39), (25, −35.28), (−108, −35.28) | main board standoffs ( **KI 3** ) |

- Measure the **1.5 mm aluminium** plate against this schedule before anything else goes on it: the pitch is **56.0** between every pair of shafts.
- With the row reading S2 → M2 left to right from the front, the calendar cluster sits **below, between N2 and K1**. A mirrored plate is scrap.

---

⚠ **Never steel.** The Hall latches read the magnets through the plate; steel would short the field away.

# 2 Glue the magnets into the dials




- Turn each `hub_dial.stl` over, back face up. The two 5 × 2 pockets sit at r 20.5, opposite the rim notch.
- Seen from the back with the notch at 12 o'clock: **N face up in the right-hand pocket, S face up in the left**. A drop of CA in each pocket.
- Same on the year dial (pockets at r 20.5) and the moon disc (r 9): seven pairs in all.

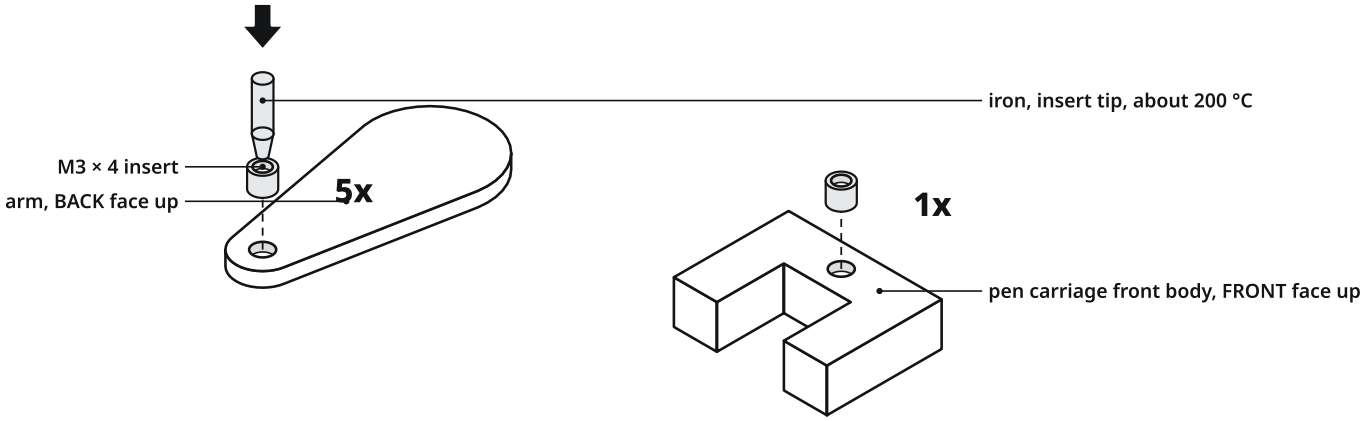
---

The latch switches where the field flips sign between the two magnets, so the edge hardly moves with gap or temperature (0.15° per mT of drift). The same order on every dial makes every station report the same edge.

---

Check each face with a spare magnet before gluing: one face pulls it, the other pushes it away.

# 3 Press in the heat-set inserts

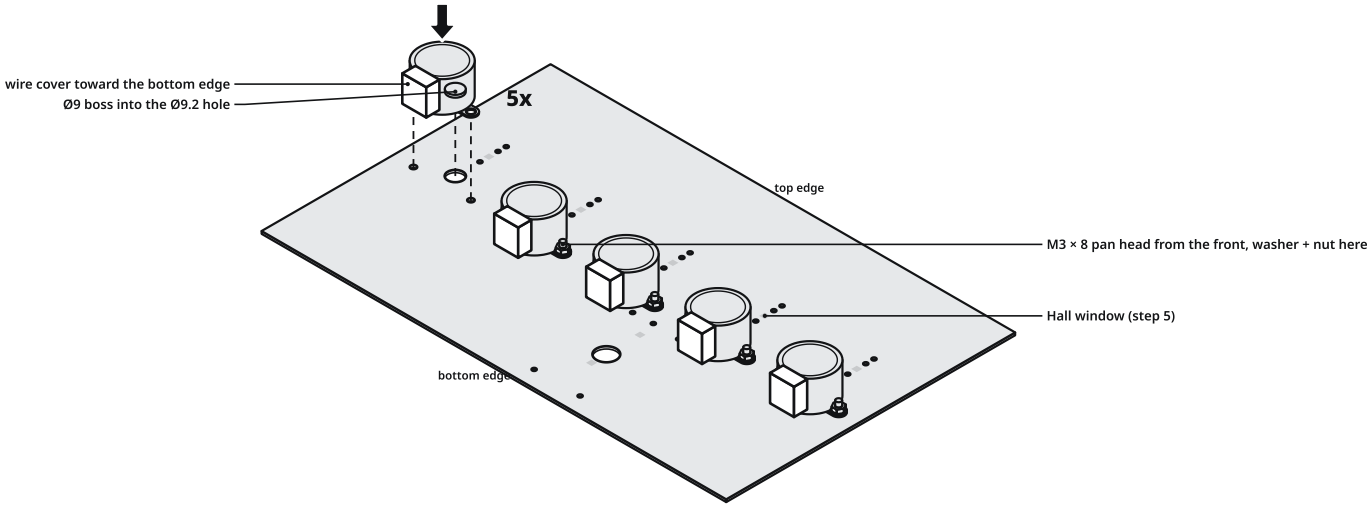



- Each of the five arms: **from the back face**, into the Ø4.0 hole under the bearing boss, flush.
- The pen carriage front body: **from the front face**, into the Ø4.0 hole at its centre. That’s the pen pulley’s screw thread (step 12).

---

⚠ Keep the insert square and stop at flush. An insert standing proud on an arm’s back face tilts the arm against the dial boss.

# 4 Bolt the motors to the back of the plate



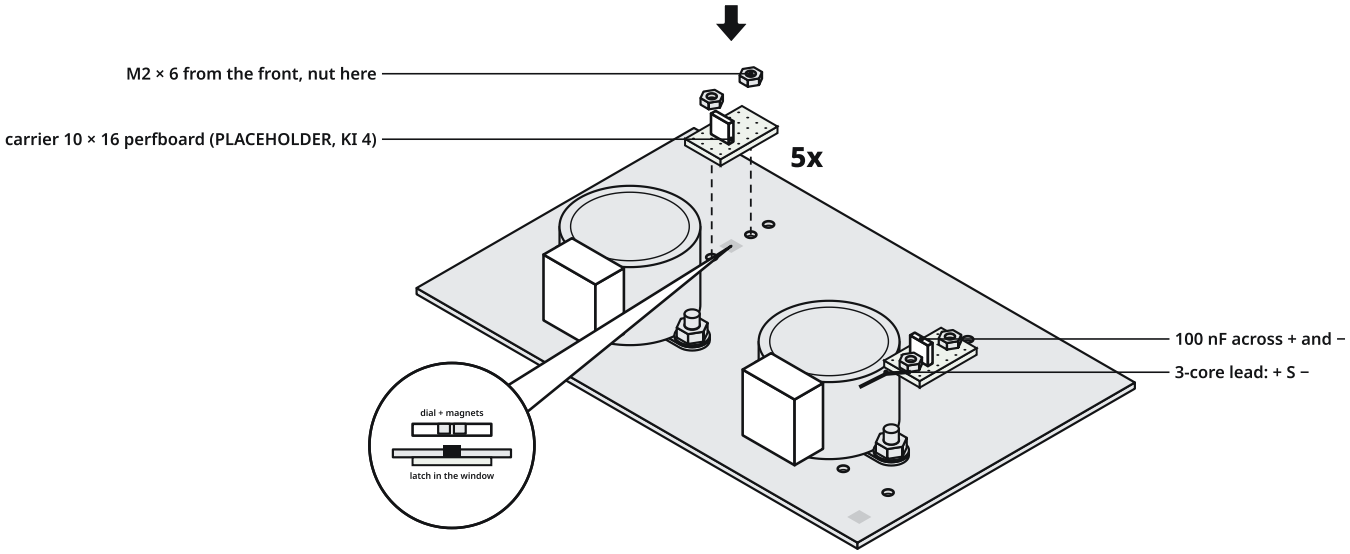

- Plate face down. Each 28BYJ-48: **boss into the Ø9.2 hole**, wire cover toward the plate's **bottom edge**.
- `M3 × 8 pan head` from the front through each ear, then a washer and a nut behind. Snug only.

---

The plate locates each shaft by its boss; the ear bolts only clamp. An M3 in the Ø4.2 ear hole would let the motor wander ±0.6 mm. The pan heads stand 2.4 proud of the front face and the dials clear them by 1.1: longer heads won't fit.

# 5 Fit the Hall carriers

KI 2 KI 4



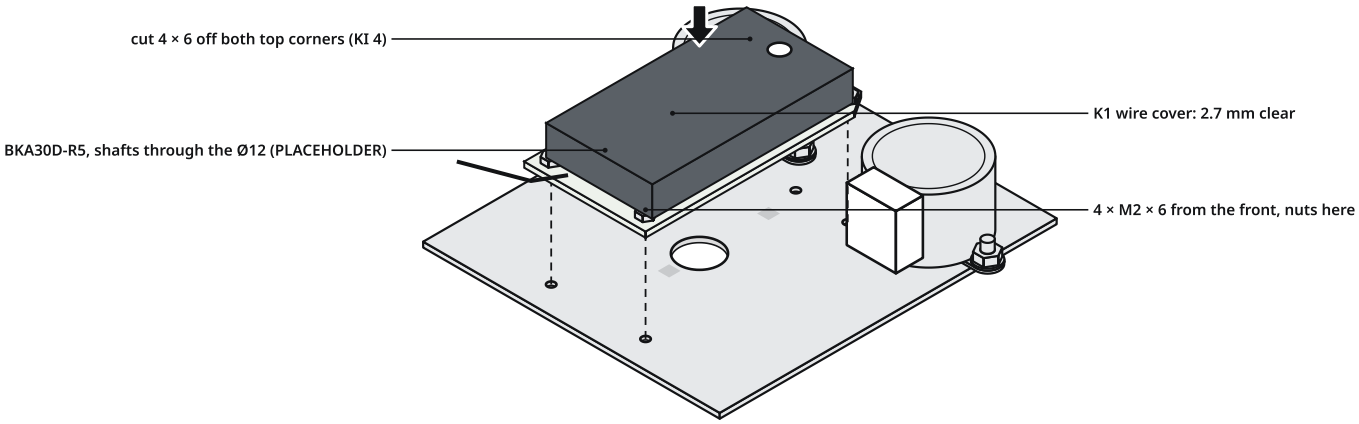

- Cut five carriers, 10 × 16, from perfboard. On each: the DRV5013 lying flat on the front face, branded face toward the plate, a 100 nF across + and −, and a 3-core lead. Label each lead with its station.
- The latch nests in its window at (x, 20.5). Two `M2 × 6` from the front through (x, 15) and (x, 26), nuts behind.

---

⚠ The DRV5013 in TO-92 has pin 1 VCC, 2 GND, 3 OUT: the carrier crosses 2 and 3 onto the lead's **+ S −** order. Check against the datasheet for your variant.

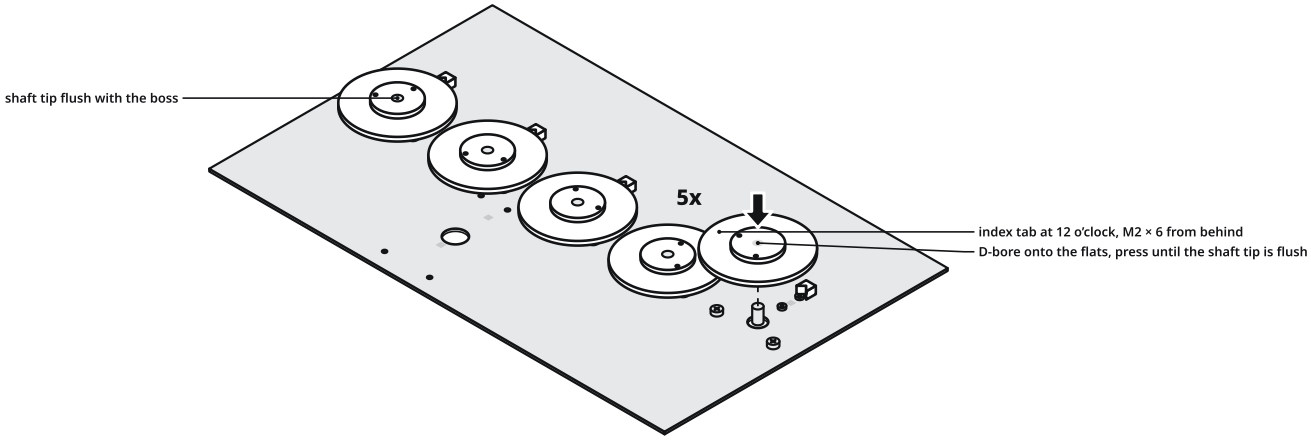
# 6 Fit the calendar carrier

KI 4 KI 5




- Cut the carrier 36 × 68 from perfboard, then **cut 4 × 6 off both top corners**. Without the cut it hits the N2 and K1 motor bodies.
- On it: the BKA30D-R5, shafts forward through the plate's Ø12, and the year and moon latches at (0, +20.5) and (0, −9) from its centre, nested in their windows.
- Four `M2 × 6` from the front, nuts behind. Then an 8-core lead for the coils (step 45), and a 3-core lead from each latch.

# 7 Press the dials onto the shafts; fit the index tabs



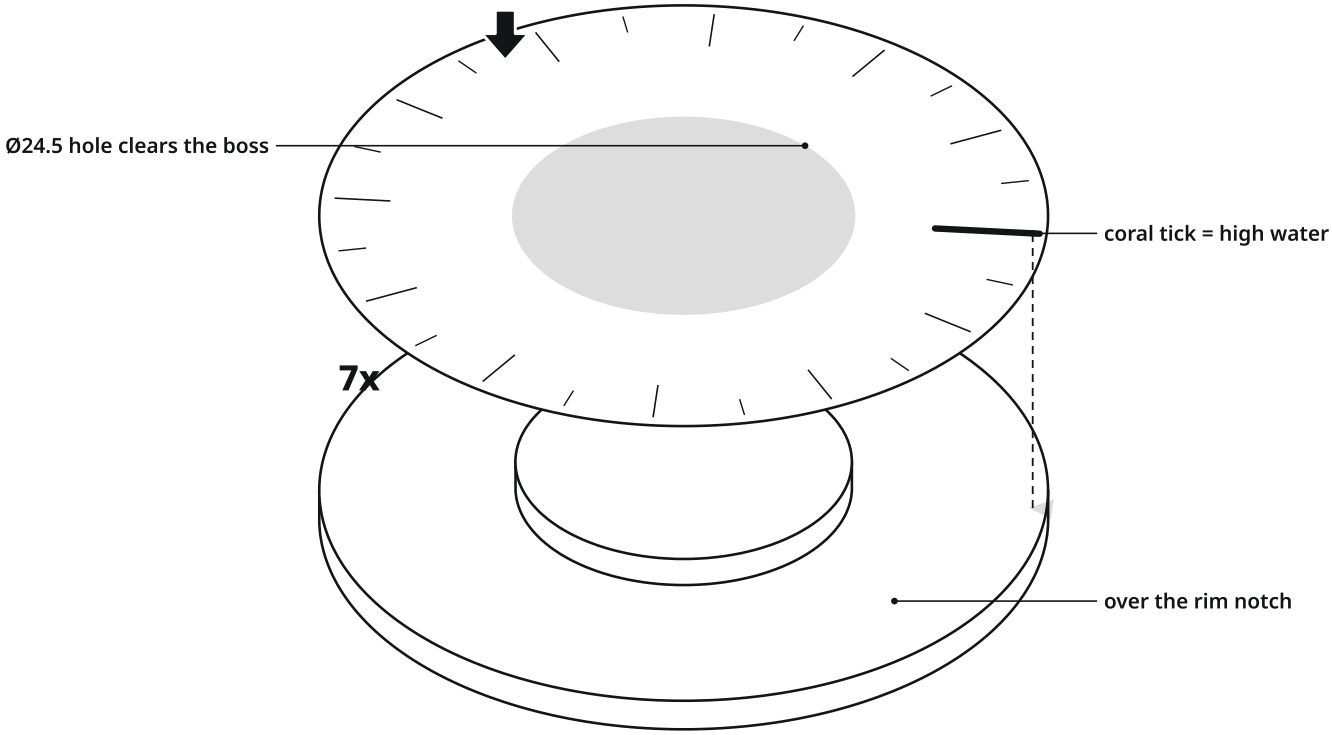

- Plate face up. Each hub + dial: the D-bore over the shaft's flats, and press **until the shaft tip is flush with the boss**. The dial's back is then 3.5 mm off the plate.
- An index tab at 12 o'clock above each dial, one `M2 × 6` from behind into its pilot.
- Turn each dial by hand: it must clear the ear-screw heads, the tab and its neighbours (4.0 mm apart).

---

ⓘ The D-bore carries the torque and a light press holds it: the line pulls in the dial's plane, so nothing pushes it off. For a loose one, a dab of clear silicone. Not threadlocker, which barely cures against plastic.

# 8 Apply the decals

KI 11



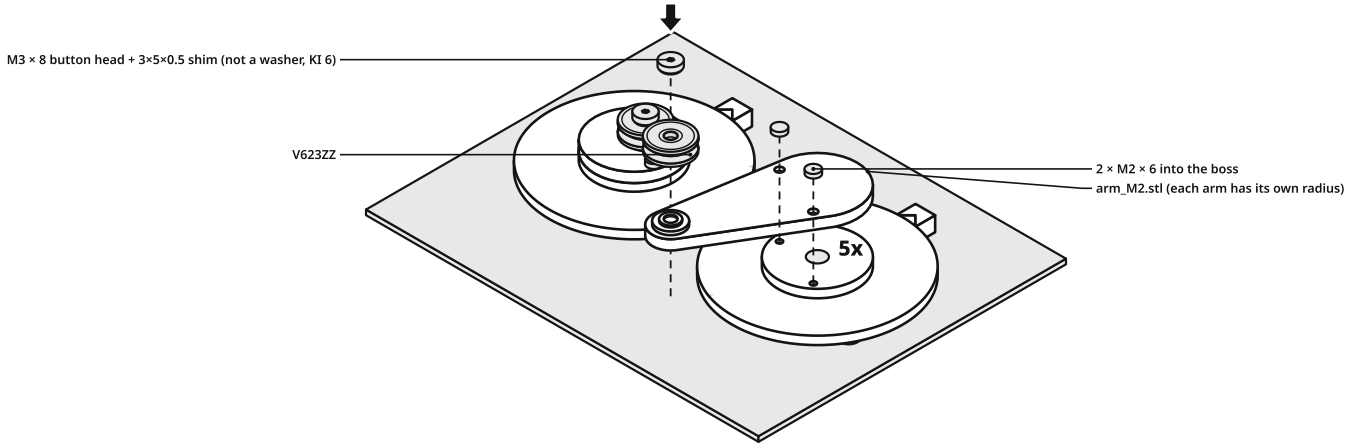

- Print `decals_1to1.pdf` at 100% on laser vinyl. Check the scale bar first.
- Each station decal (Ø52, Ø24.5 hole): **coral tick over the rim notch**. Year decal on the year dial, moon decal on the moon disc.

---

The scales are warped: each tick sits where the arm really is (M2's ticks move up to ±33 min). The decals depend on the constituents, the longitude and the year, so they're reprinted for each deployment.

# 9 Bolt on the arms and their bearings

KI 6



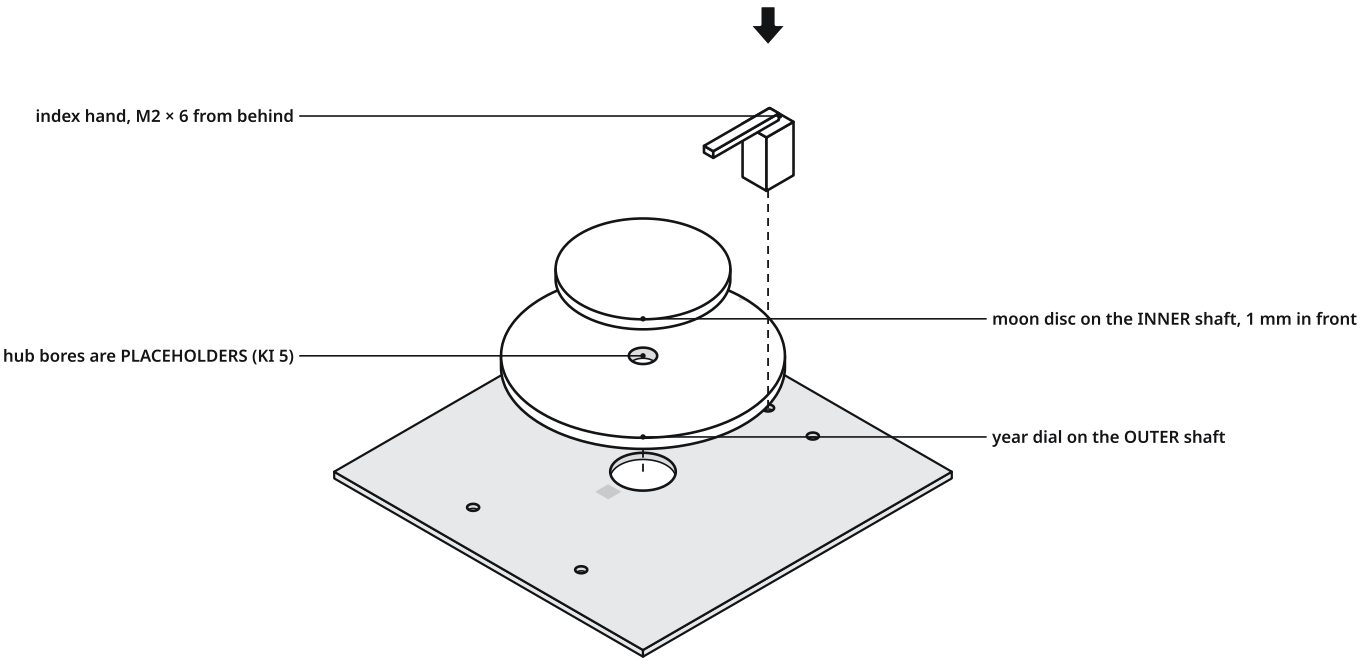

- Each arm onto its own dial: `arm_S2` … `arm_M2`, stamped by radius. Two `M2 × 6` into the boss's pilots.
- A `V623ZZ` onto the stepped boss, a **3 × 5 × 0.5 shim**, then an `M3 × 8 button head` into the insert. Firm: the inner race is clamped, the outer spins.
- Spin each bearing: free, no scrape. M2's arm passes 2.5 mm over O1's dial.

---

When real constituents replace the placeholders, only the arms are reprinted: the pin radius is the amplitude (35 mm per metre of tide).

# 10 Fit the year dial, the moon disc and the index hand

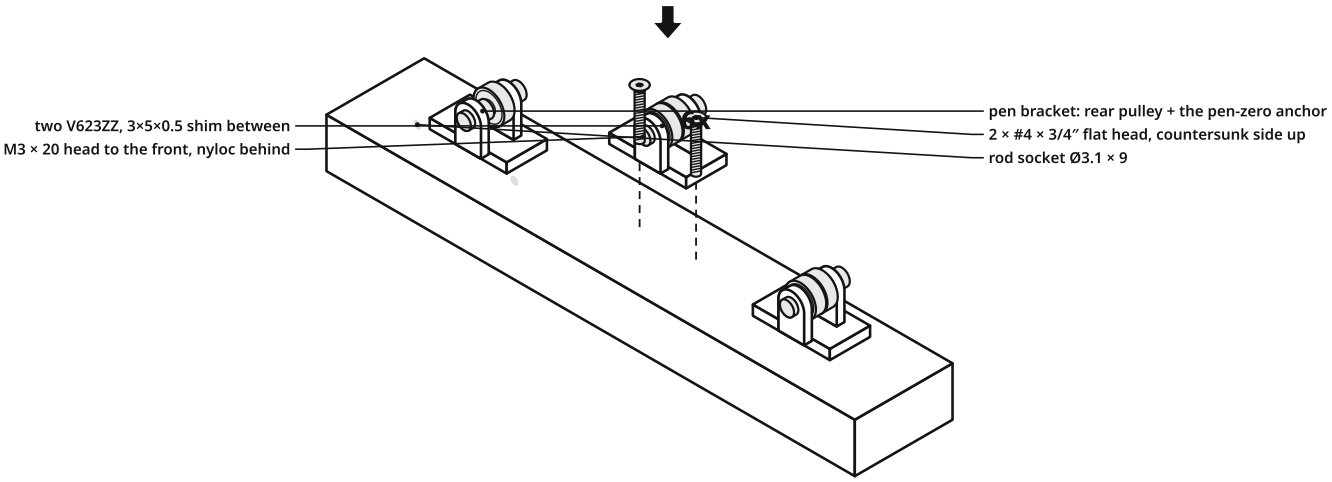
KI 5




- Year dial onto the BKA30D-R5's **outer** shaft, magnets to the plate. Moon disc onto the **inner** shaft, 1 mm in front of the year dial.
- The index hand, one `M2 × 6` from behind: it crosses the year ring and stops at the moon disc's rim.
- Turn both by hand: the gauge motor is weak (a 2.5 g pointer rating), so the discs must spin with no rub at all.

# 11 Screw the pulley brackets to the top rail

KI 13



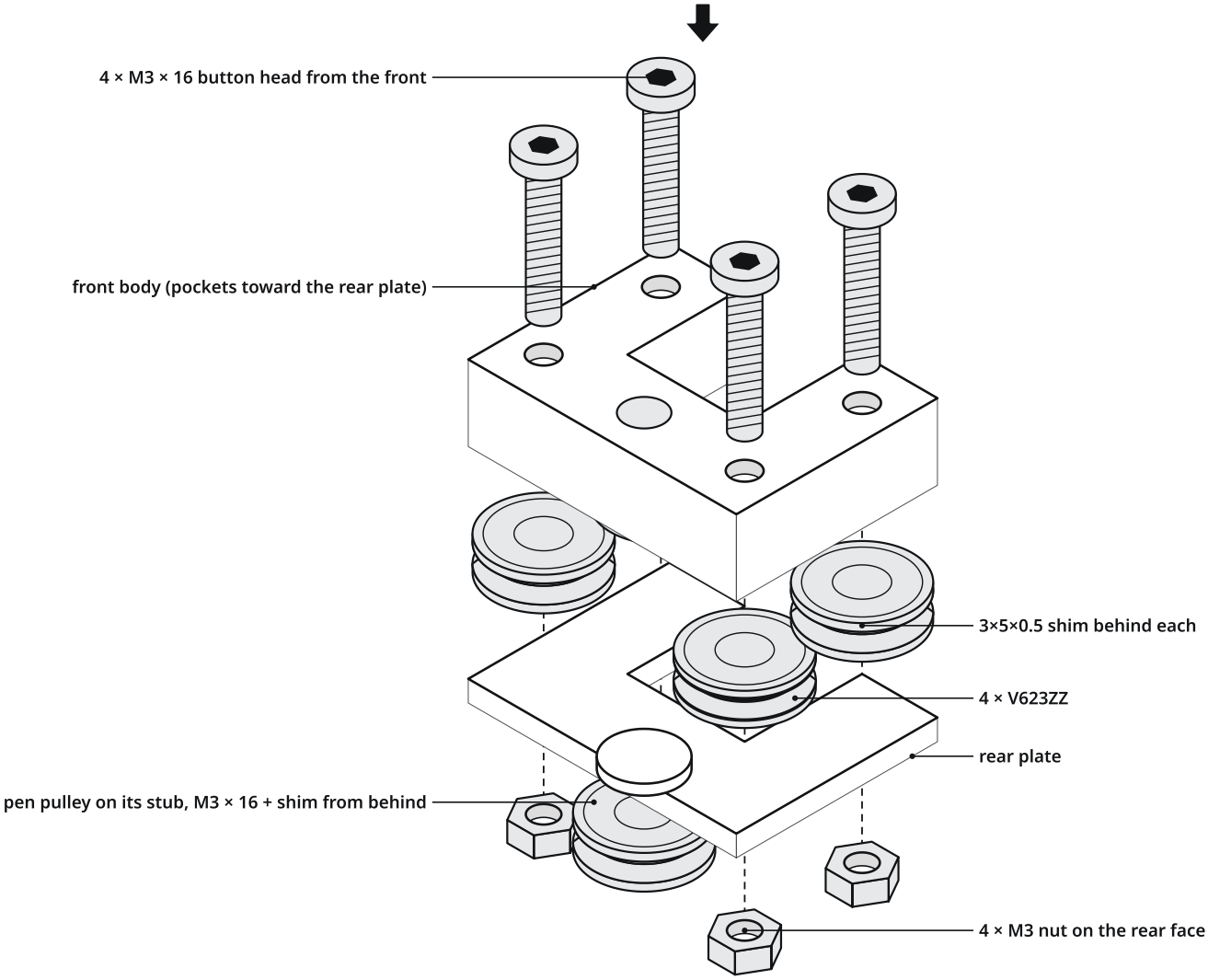

- Each bracket: two `V623ZZ` with a **3 × 5 × 0.5 shim** between them on an `M3 × 20`, head to the front, nyloc behind. Snug: they must spin freely.
- Five brackets directly above the shafts (x −224 … 0) and the pen bracket at x +70. `#4 × 3/4″` into pilots at station x ± 11, 15 from the rail's rear edge; the notch on each bracket marks its station and faces front.
- The pen bracket carries one pulley (rear) and the screw-adjustable anchor (front).

---

ⓘ The two pulleys run in planes 4.5 apart, either side of the arm-bearing groove at z −15. That's what lets the threading mirror at alternate stations (step 16).

# 12 Build the pen carriage

KI 6 KI 7



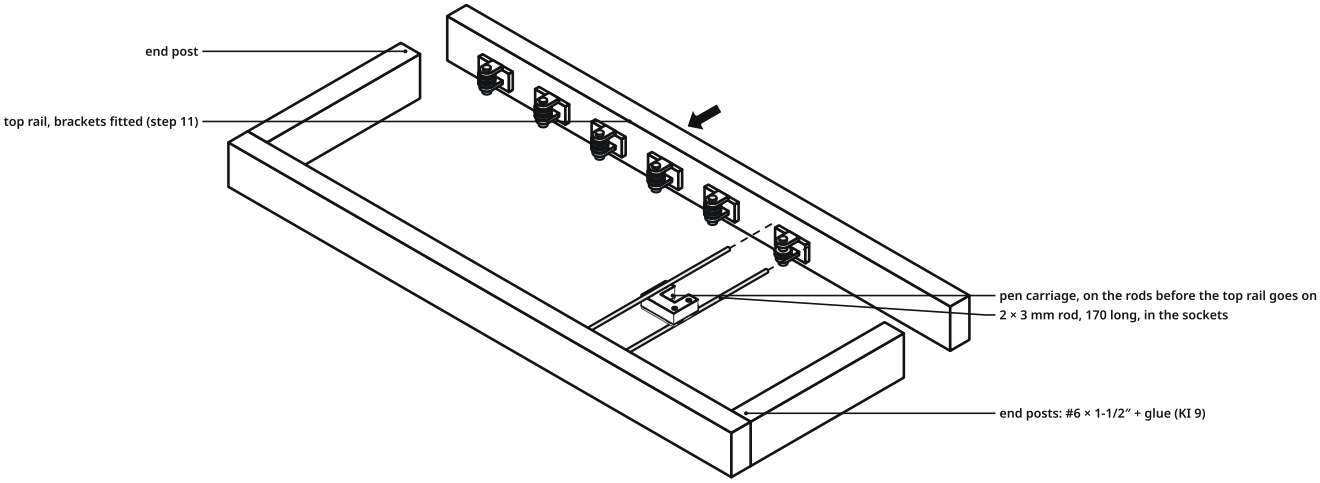

- Pen pulley on the rear plate's stub: `M3 × 16` with a shim, from behind, into the front body's insert.
- Four bearings, each with a shim on its rear side, clamped between rear plate and front body by `M3 × 16` from the front and a nut behind.
- Leave the +x pair (slotted holes) loose until the carriage is on its rods (step 13).

---

ⓘ The inner races are the spacers. The V grooves locate the carriage front to back; the two sides pushing outward on opposite rods locate it sideways.

# 13 Close the frame around the rods and the carriage

KI 1 KI 9 KI 13



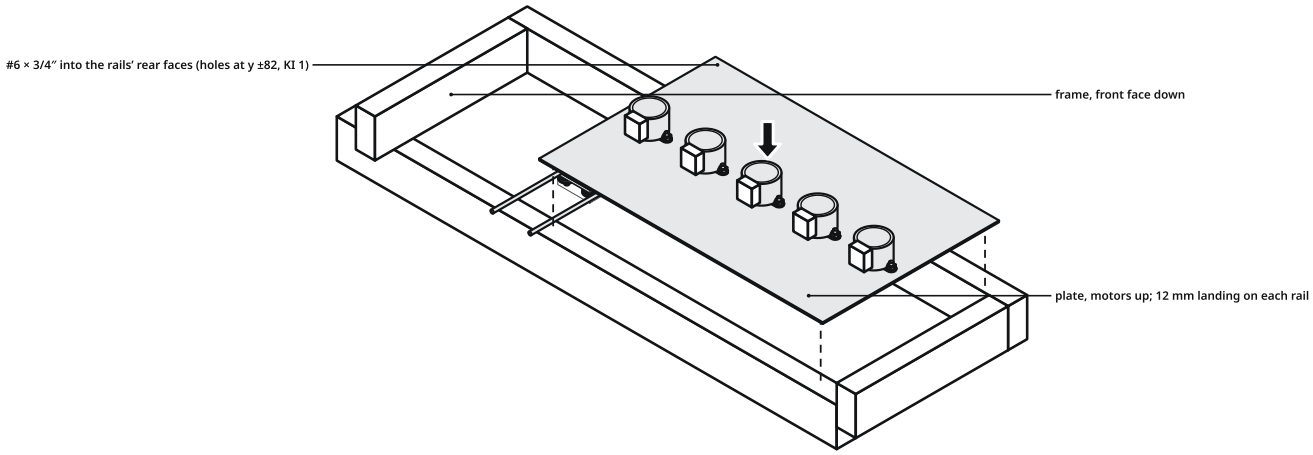

- Tape the 1:1 templates to the rails (KI 13) and drill them as a mirrored pair, clamped outer face to outer face: rod sockets **Ø3.1 × 9** at 325.08 and 362.92 from the S2 end, 23.75 from the rear edge; the drum hole at 429.45, 24 from the rear edge.
- Bottom rail flat, both rods in its sockets, the carriage between them with its bearings on the rods. Then the top rail onto the rod ends.
- End posts between the rails, `#6 × 1-1/2″` and glue. Push the +x bearing pair outward for a light preload and lock the two screws.

---

⚠ The rods are captive once the top rail is on: the carriage must go on them first.

# 14 Screw the faceplate to the frame

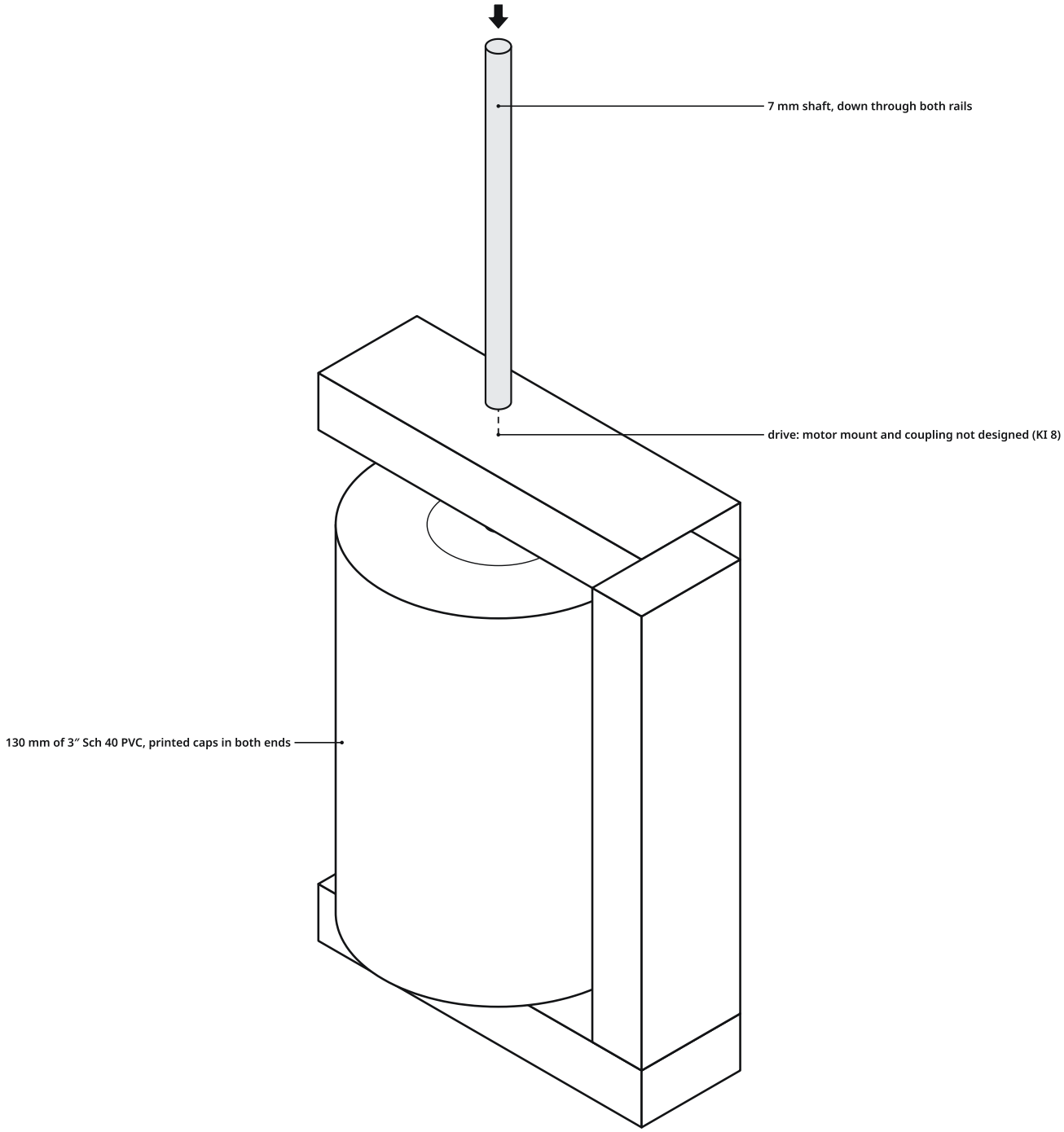
KI 1




- Frame face down. The plate comes down onto the rails' rear faces, motors up, with **12 mm of plate landing on each rail**.
- `#6 × 3/4″` through the Ø4.0 holes into the rails. Then stand the machine up and check every arm bearing clears the pulleys above it (18 mm at M2).

# 15 Fit the drum

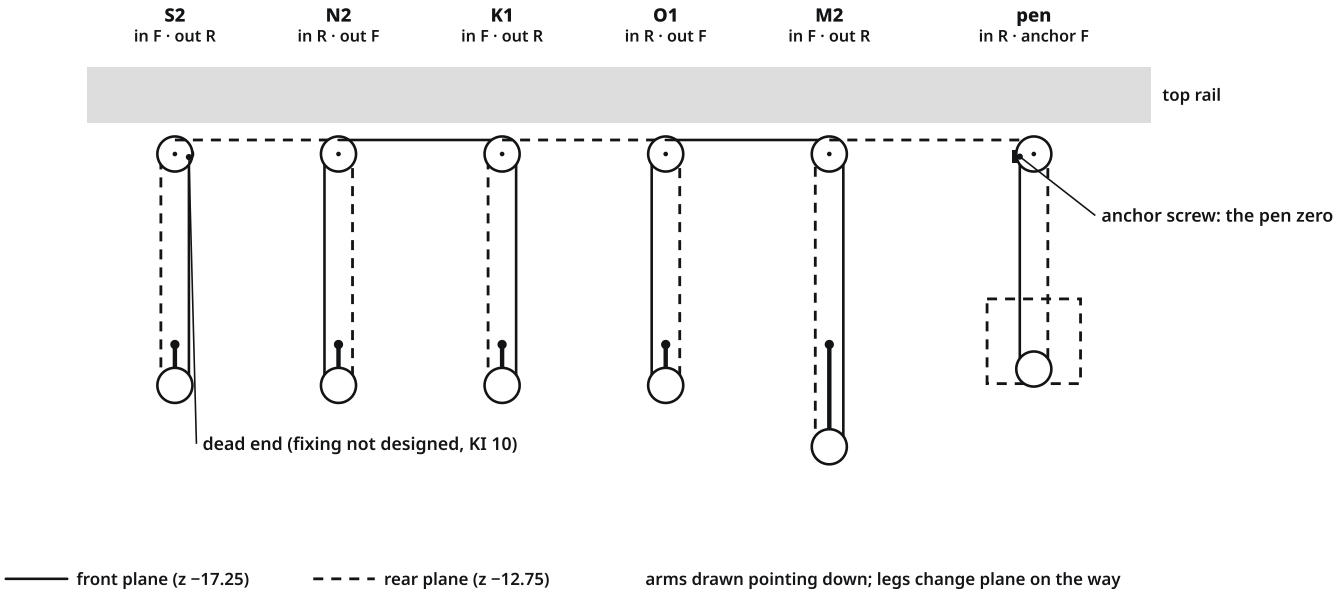
KI 8




- Cut 130 mm of 3″ PVC square; press an end cap into each end.
- The 7 mm shaft down through the top rail, the drum and into the bottom rail's plain bearing, with a washer under the bottom cap.
- The drum axis sits at x 155.5, z −24, on the carriage's mid-plane; 5.95 mm clear of the right post.

# 16 Thread the line

KI 10




| | S2 | N2 | K1 | O1 | M2 | pen |
|---|---|---|---|---|---|---|
| in over | front | rear | front | rear | front | rear |
| out over | rear | front | rear | front | rear | front = anchor |

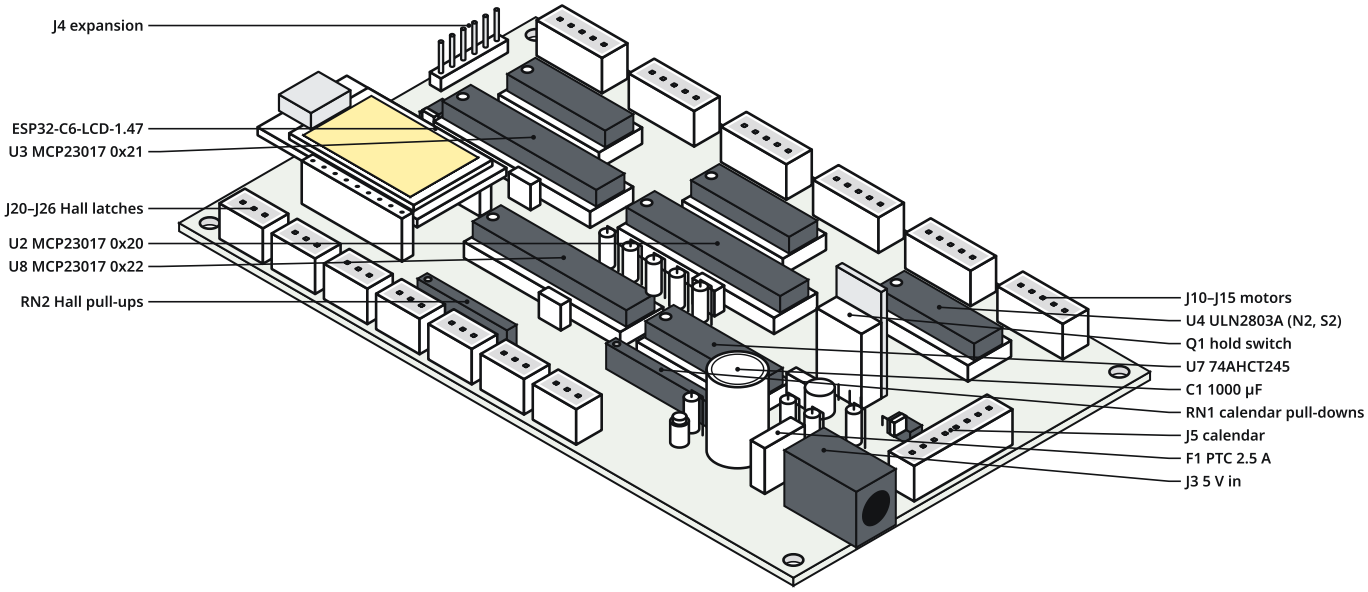
- Crimp a loop in the far end of the **0.30 mm fluorocarbon** and fix it at S2's bracket, front plane.
- At each station: over the IN pulley, down the +x side, **round under the arm's bearing**, up the −x side, over the OUT pulley. Then along the rail to the next one.
- At the pen: over the rear pulley, round under the carriage's pen pulley, up to the anchor screw on the front plane. Crimp it there.

ⓘ The order mirrors at every station, so every run between stations stays in one plane and only the legs skew (about 2°). The line passes **round** each arm bearing and never attaches: the U-turn turns a rise d into 2d of take-up, and the pen pulley halves it again, so the pen moves 35 mm per metre of tide.

# B Main board

140 × 81.3 mm, all through-hole, chips and the ESP32 in sockets. Written for a first board: one kind of part at a time, simplest first, each with a map of where it goes. Then test everything (section C) before any chip or the C6 goes in.

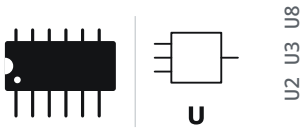
## The finished board




Motor sockets across the top edge, Hall sockets along the bottom, calendar and power on the right, the ESP32-C6-LCD-1.47 on the left with its USB-C out of the left edge. Nothing stands under the C6.

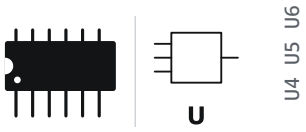
# 4

## Electronic parts for the board



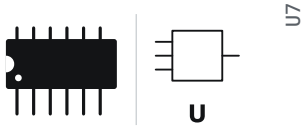

**3x**

**MCP23017-E/SP**, DIP-28, + socket: I/O expanders 0x20, 0x21, 0x22



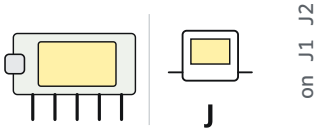

**3x**

**ULN2803A**, DIP-18, + socket: motor drivers, two motors each



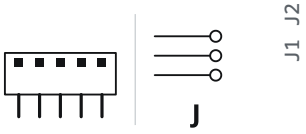

**1x**

**SN74AHCT245N**, DIP-20, + socket: calendar coils at 5 V



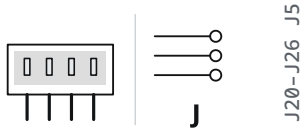

**1x**

**Waveshare ESP32-C6-LCD-1.47**: plugs in, not soldered



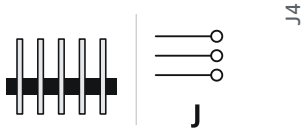

**2x**

**1 × 9 female header**, 8.5 tall: the C6's socket



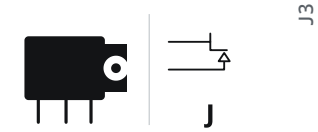

**14x**

**JST-XH** vertical: B5B ×6 (motors), B3B ×7 (Halls), B8B ×1 (calendar)



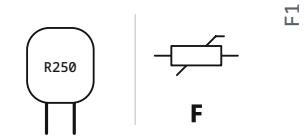

**1x**

**1 × 6 male header**: expansion (3V3 GND SDA SCL IO1 IO3)



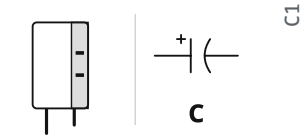

**1x**

**PJ-102AH** barrel jack, 5.5 / 2.0 mm



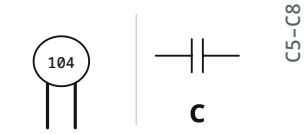

**1x**

**MF-R250** resettable fuse, 2.5 A hold



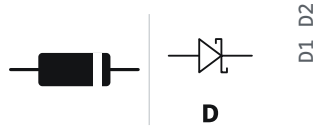

**1x**

**1000 µF** 10 V or more, radial 10 mm



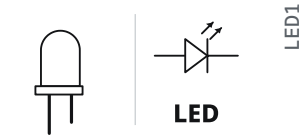

**4x**

**100 nF** ceramic, 2.5 mm pitch (7 more on the Hall carriers)



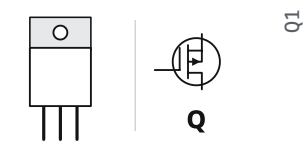

**2x**

**1N5819** Schottky: C6 feed, and the hold switch's freewheel



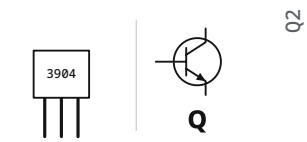

**1x**

**3 mm green**: power



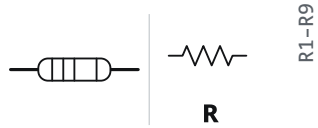

**1x**

**IRLIB9343** P-FET, TO-220: switches the motors' 5 V



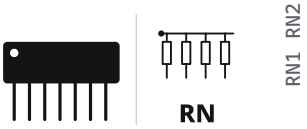

**1x**

**2N3904**, TO-92: drives Q1's gate




**9x**

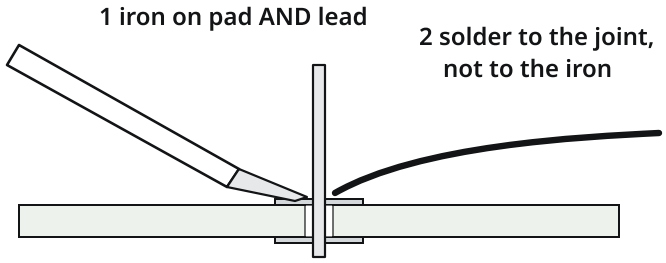
**¼ W, standing**: 4.7k R1 R2 · 10k R3 R4 R7 R9 · 1k R5 R6 R8

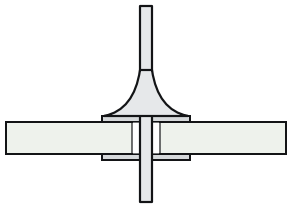



**2x**

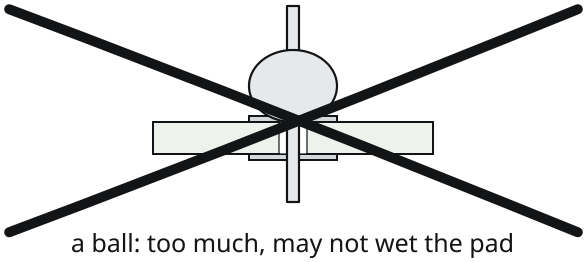
**Bussed 10k SIP**: RN1 9-pin (4609X), RN2 8-pin (4608X)

# 17 Soldering, the basics

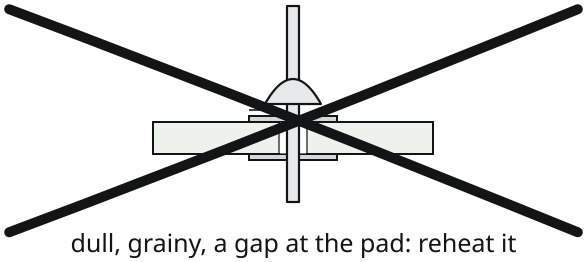


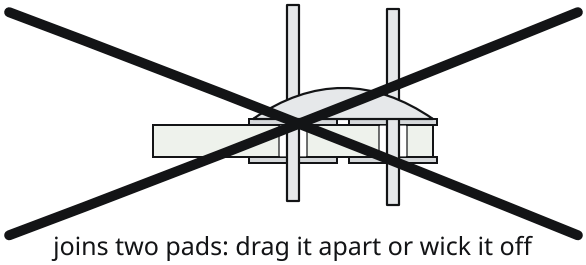
a shiny cone that hugs the lead



a ball: too much, may not wet the pad



dull, grainy, a gap at the pad: reheat it



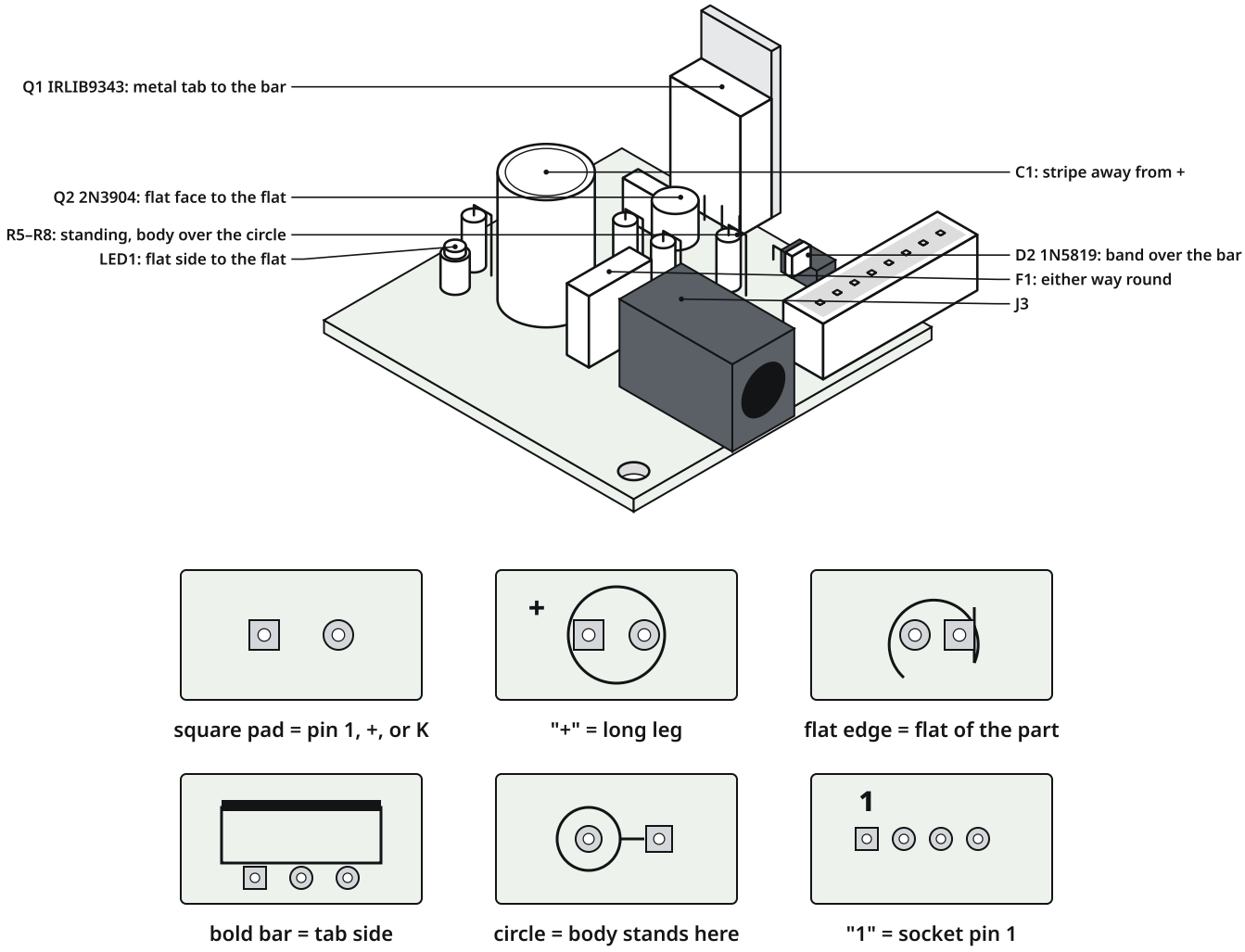
joins two pads: drag it apart or wick it off

- Iron at **330 to 350 °C**. Wipe the tip on damp sponge or brass wool, then melt a little solder onto it: a shiny tip moves heat.
- **0.8 mm rosin-core solder**. 63/37 leaded flows most easily; wash your hands after. Work with a fan or open window.
- Parts go in from the **printed side**. Solder from the back. Trim each lead just above the cone, after it is soldered.

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Never soldered before? Practise ten joints on a scrap of perfboard first. The Hall carriers are perfboard too, so the practice pays twice.

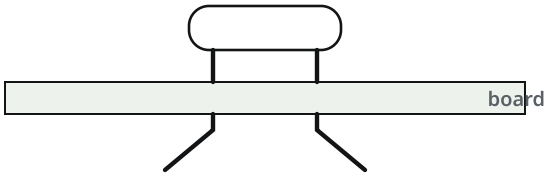
# 18 Read the silkscreen




- The **square pad** is the one to find: pin 1 of a socket or connector, **+** of C1, the band end of a diode, **K** of the LED.
- Hold the board as the maps show it: **motor sockets along the top**, Hall sockets along the bottom, the C6 on the left.
- Every part here has one right way round except the resistors, the ceramic capacitors and F1. If a part does not match its outline, stop and check.

# 19 The order of work

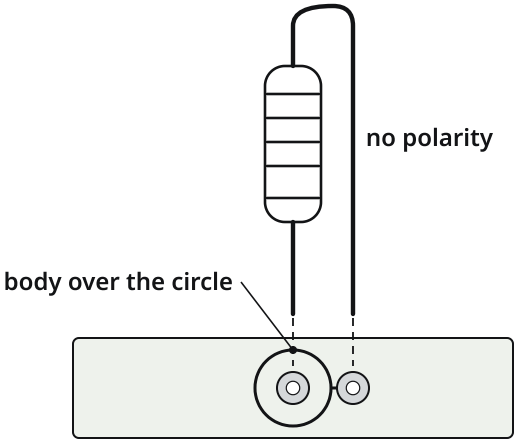
| | Kind | How many | New idea |
|---|---|---|---|
| 1 | Resistors | 9 | bending a lead, reading a value |
| 2 | Resistor networks | 2 | a dot marks pin 1 |
| 3 | Ceramic capacitors | 4 | reading a code |
| 4 | The big capacitor, C1 | 1 | **polarity**: one way round only |
| 5 | Diodes and the LED | 3 | a band or a flat marks the end |
| 6 | Chip sockets | 7 | a notch marks pin 1 |
| 7 | C6 sockets and the expansion header | 3 | female and male |
| 8 | JST-XH connectors | 14 | the housing's open side |
| 9 | Fuse and power jack | 2 | big pins, big joints |
| 10 | Transistors | 2 | reading the part number, the tab |




**splay the legs under the board so the part cannot fall out**

- Fit all of one kind, splay the legs, turn the board over, solder them all, trim them all, tick them off on the map.
- When kind 10 is done, the quick check (step 32) lights the board for the first time. Then section C tests it properly, **before** any chip or the C6 goes in.

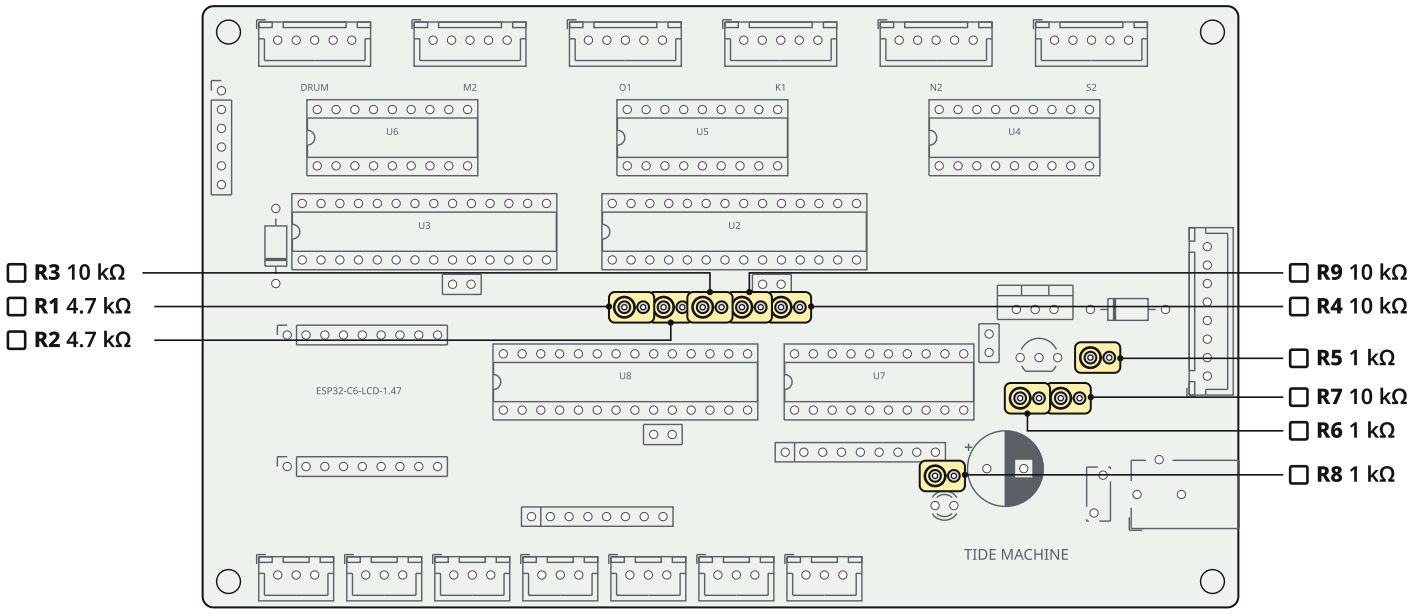
# 20 Resistors: how




- These resistors **stand up**. Bend one lead tightly over the top of the body and down beside it. The body stands over the **circle**; the bent lead goes into the other hole.
- No polarity: either end can go up.
- **Measure every resistor** with the meter on Ω before it goes in. The colours below confirm it.

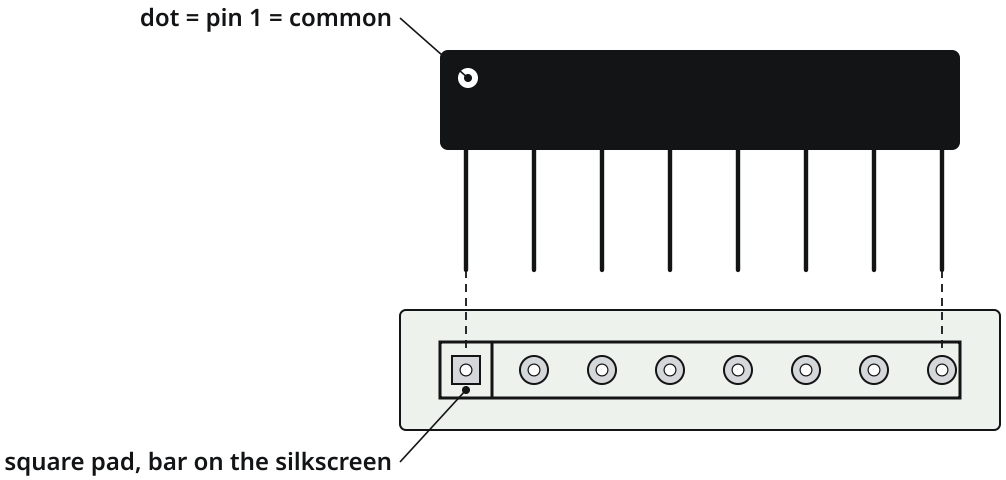
| | ordinary (4 bands, 5 %) | precision (5 bands, 1 %) |
|---|---|---|
| **10 kΩ** |  brown · black · orange · gold |  brown · black · black · red · brown |
| **4.7 kΩ** |  yellow · violet · red · gold |  yellow · violet · black · brown · brown |
| **1 kΩ** |  brown · black · red · gold |  brown · black · black · brown · brown |

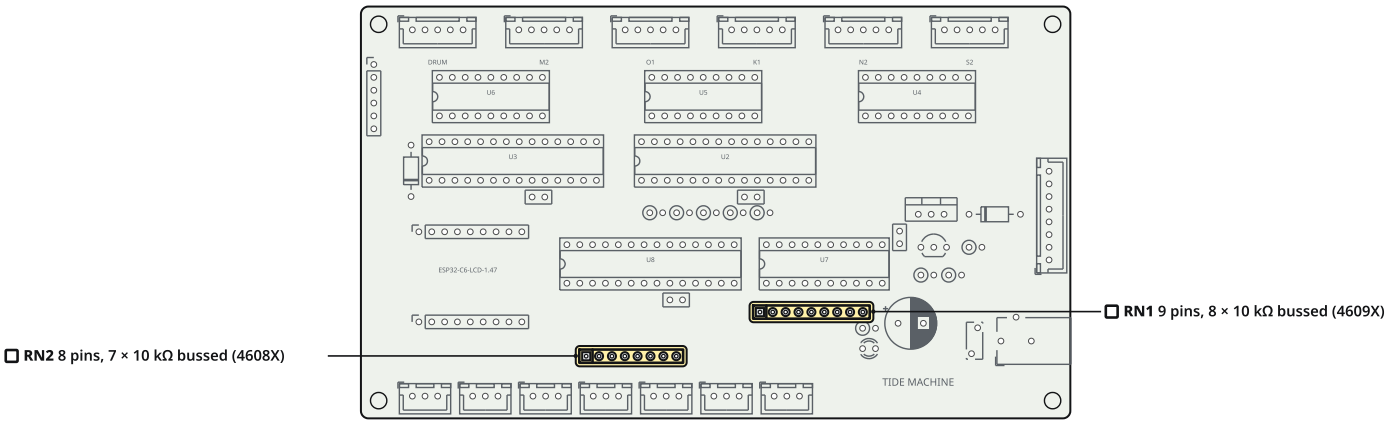
# 21 Resistors: where




| Value | How many | Where |
|---|---|---|
| 10 kΩ | 4 | R3 R4 R7 R9 |
| 4.7 kΩ | 2 | R1 R2 (the I²C pull-ups) |
| 1 kΩ | 3 | R5 R6 R8 |

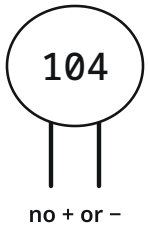
# 22 Resistor networks: how and where



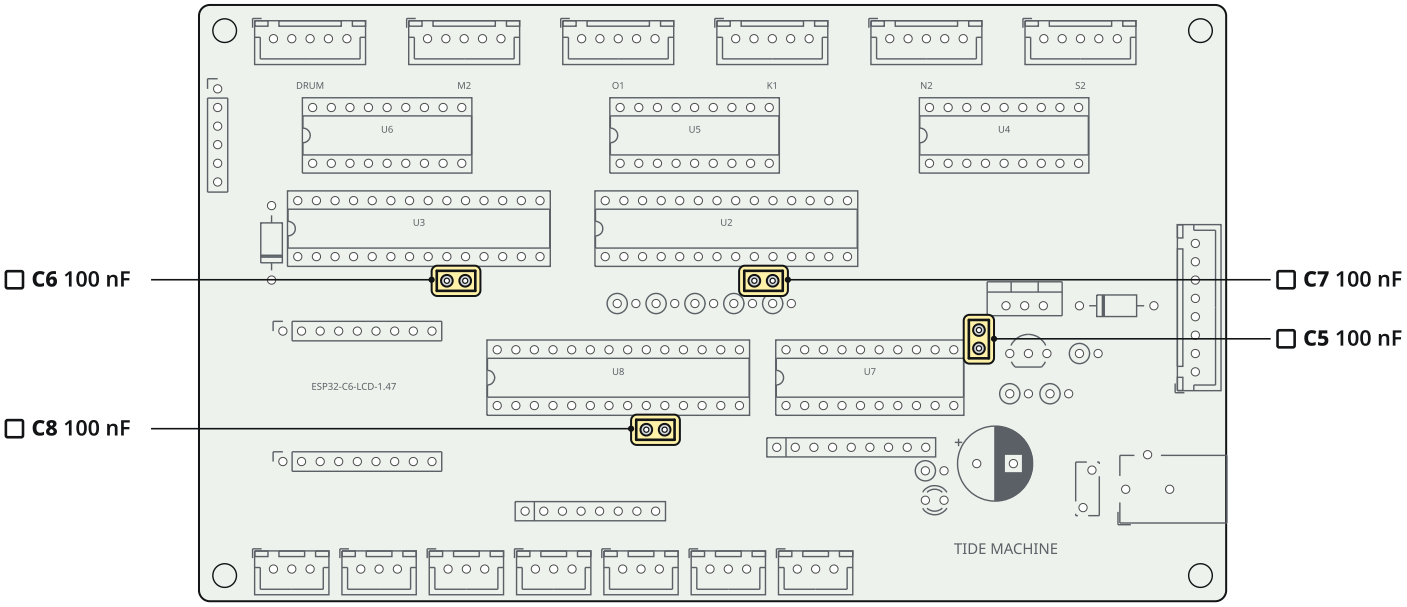




- A row of resistors in one flat package. The **dot** printed on it is pin 1, the common pin: it goes in the **square pad**.
- RN1 has 9 pins and RN2 has 8. They only fit their own place, but they fit it backwards too: check the dot.

# 23 Ceramic capacitors: what and where

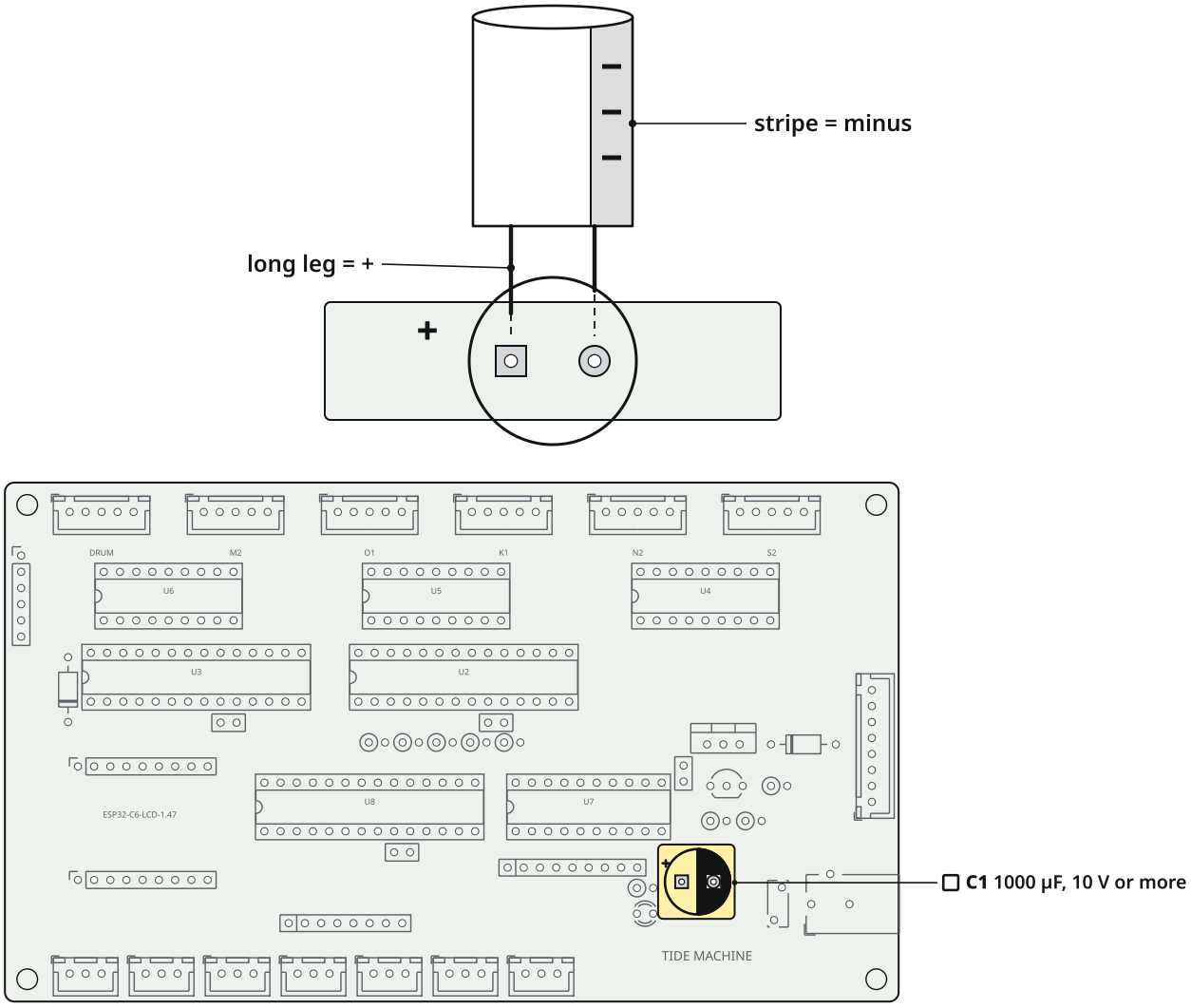



| | |
|---|---|
| `10 then 4 zeros` | the first two digits, then a count of zeros |
| `100 000 pF` | picofarads |
| `= 100 nF` | nanofarads: the value in the parts list |
| `= 0.1 µF` | microfarads: how some sellers list it |




- **Four, all the same: 100 nF, printed 104.** No polarity. Seven more of the same go on the Hall carriers later.

# 24 The big capacitor, C1: how and where

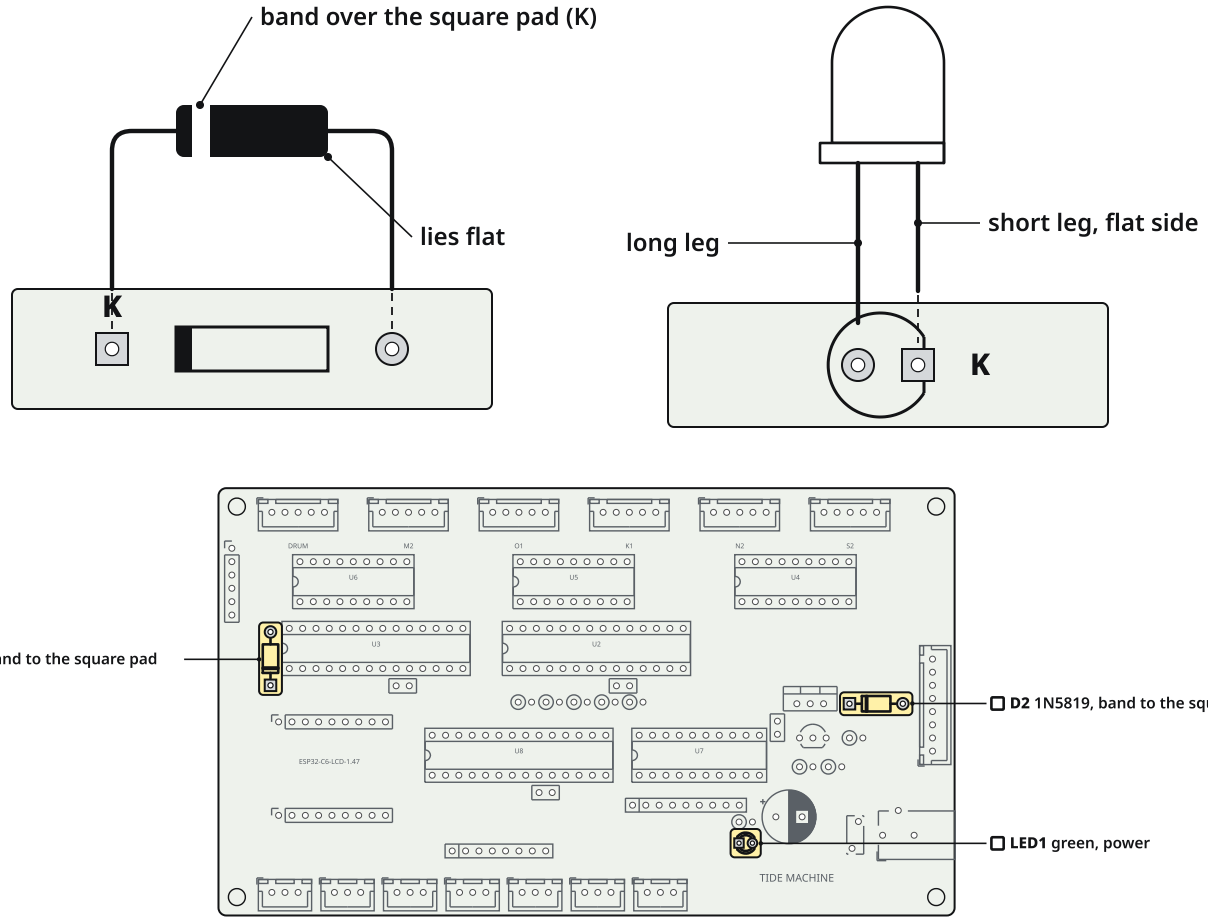



- **Polarity**, the first part on the board that has it: the **long leg is +**, the **stripe** down the sleeve marks **−**. Long leg into the **square pad**, beside the printed **+**.

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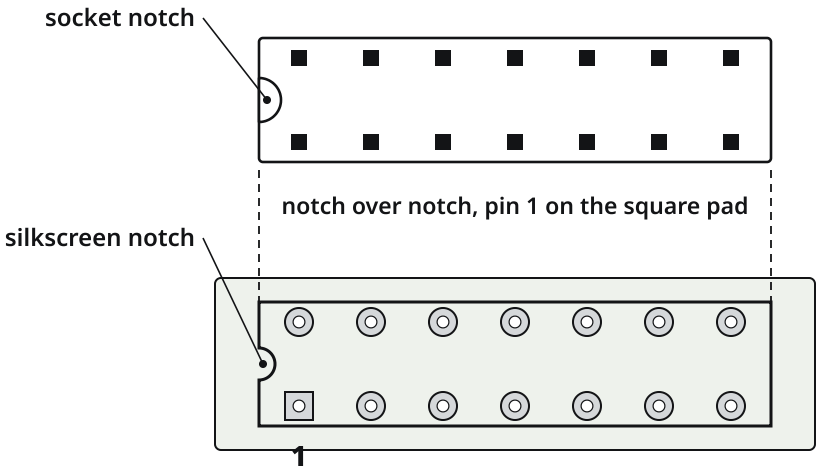
⚠ Fitted backwards, an electrolytic can swell or burst when power goes on. A higher voltage rating than 10 V is fine; a lower one is not.

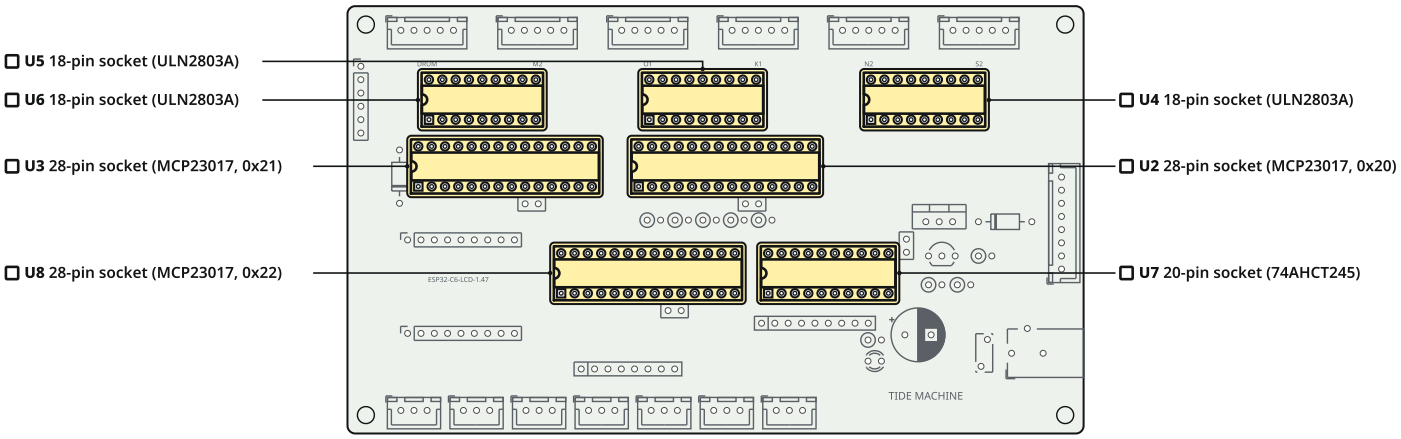
# 25 Diodes and the LED: how and where




- **D1, D2** (1N5819) lie **flat**, with the **band** over the **square pad**.
- **LED1:** the short leg and the flat side of the rim go to the square pad.

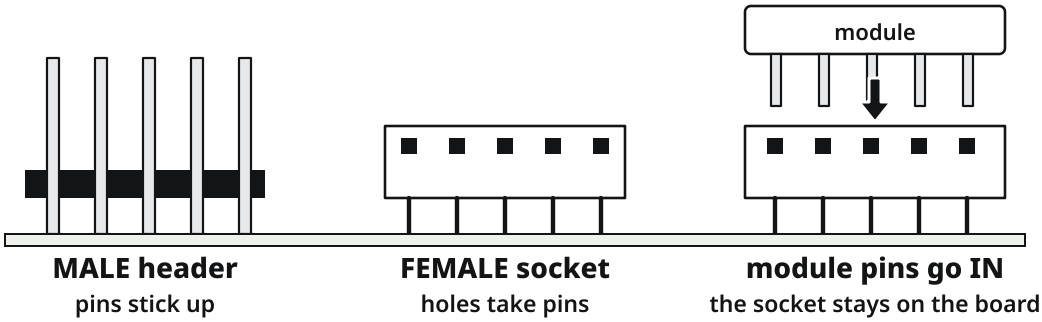
# 26 Chip sockets: how and where



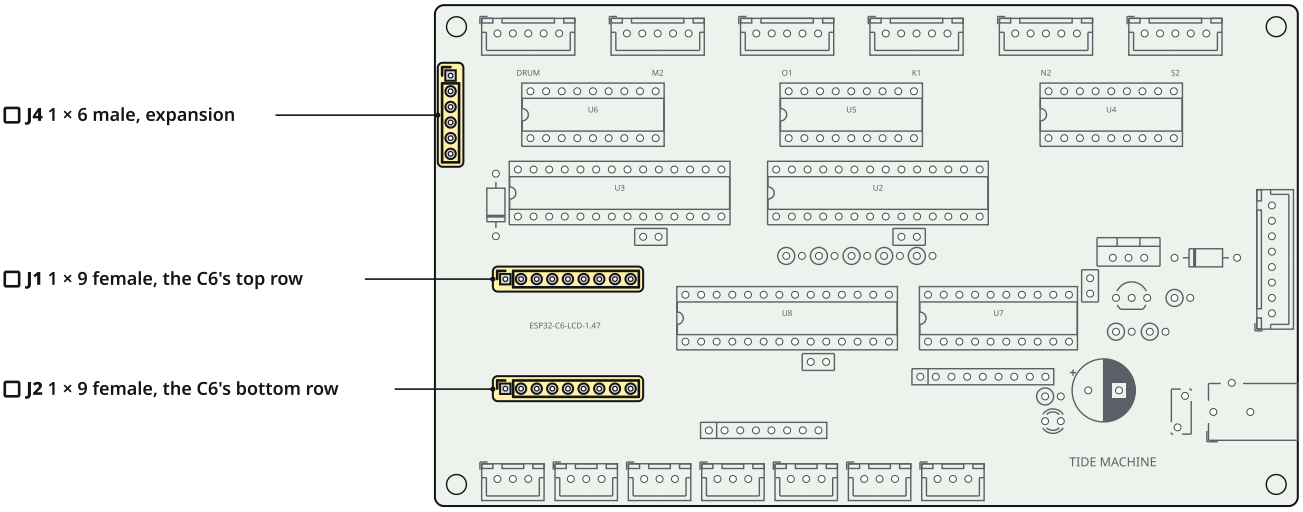

- Sockets only: **no chips yet**. They go in at step 40, after the tests.
- The socket's **notch** sits over the notch printed on the board. Solder two opposite corners, check it sits flat, then the rest.

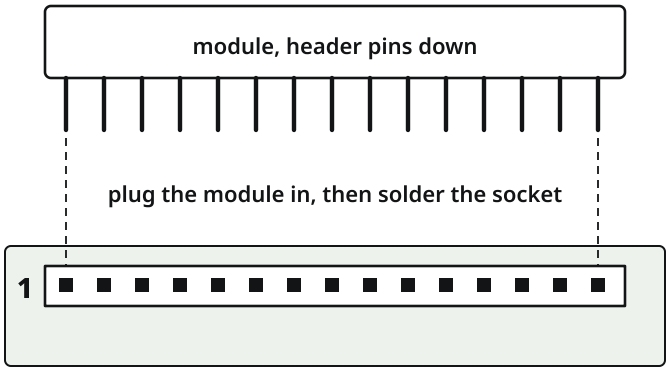
# 27 Male and female




- A **male** header has **pins**. A **female** socket has **holes**.
- **J1, J2** are **female**: the C6 plugs into them. **J4** is **male**: a cable plugs onto it later.
- The JST-XH connectors (step 29) are male too: pins inside a little plastic wall.

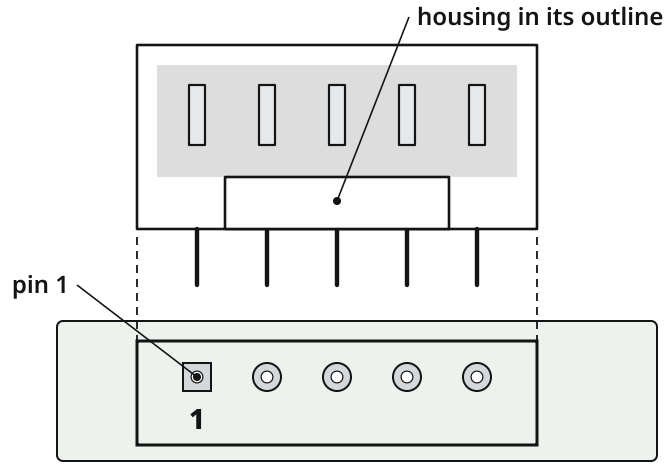
# 28 C6 sockets and the expansion header: how and where

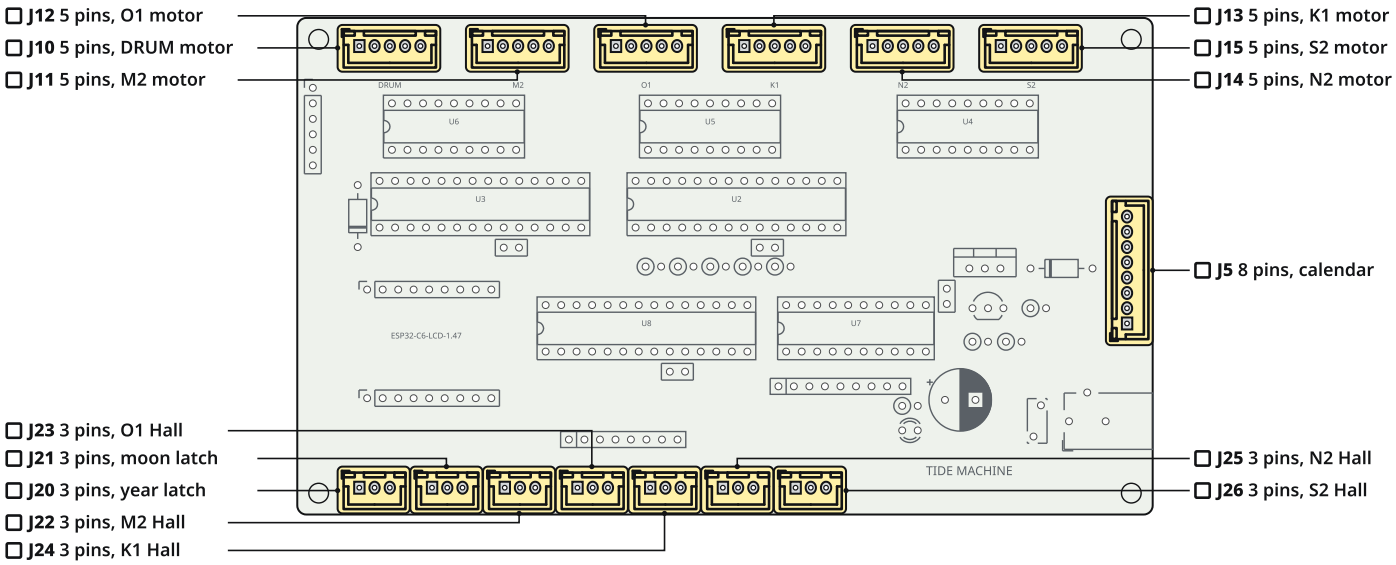



- **J1, J2:** plug both onto the C6's pins, set the C6 in the board, turn it over and solder. Then **unplug the C6** again: it goes in at step 39.
- **J4:** short ends of the pins down. Solder one pin, check it stands upright, then the rest.

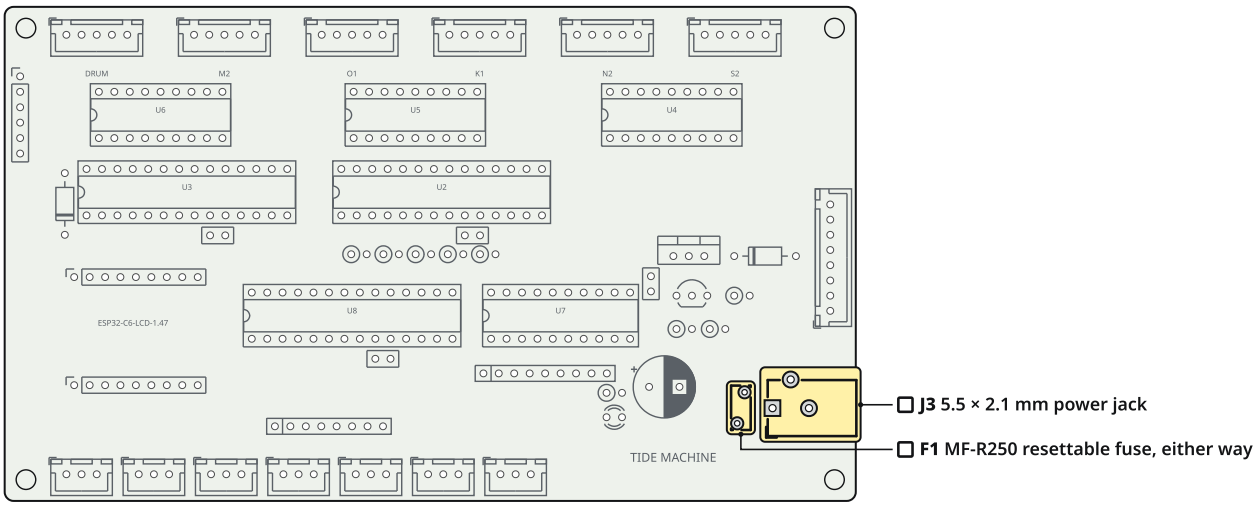
# 29 JST-XH connectors: how and where



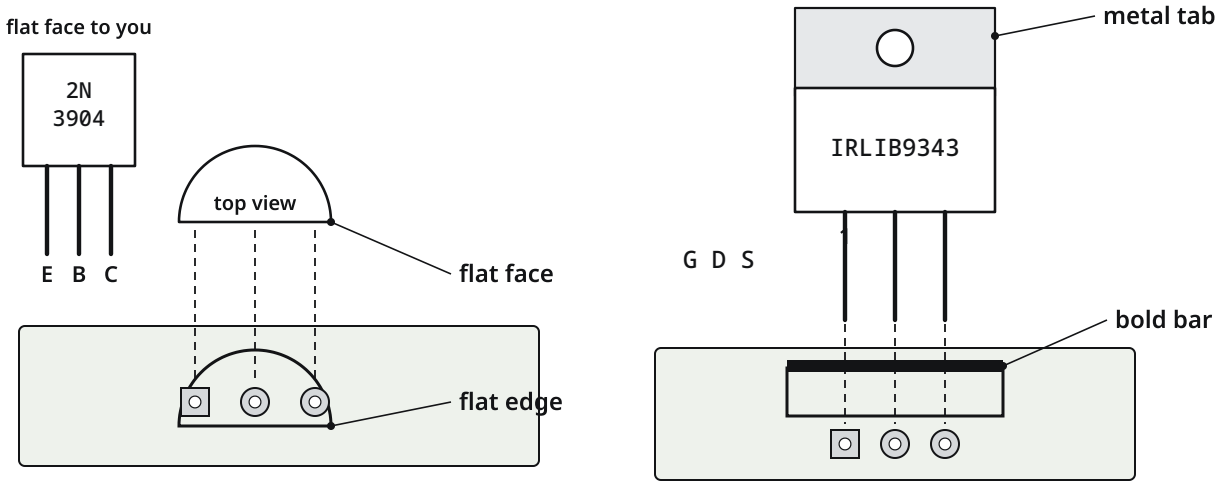

- Six with 5 pins (motors, top edge), seven with 3 pins (Halls, bottom edge), one with 8 pins (calendar, right edge).
- The housing sits **inside its printed outline**, pin 1 on the square pad. Both ways fit the holes; only that one is right.

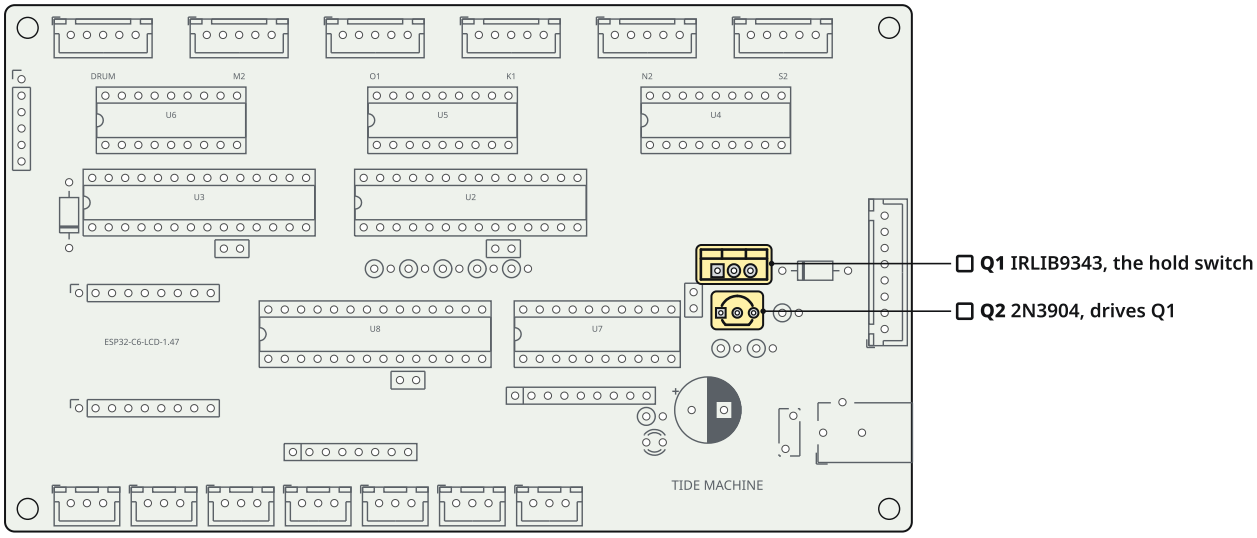
# 30 Fuse and power jack: how and where




- **F1**, the resettable fuse, fits either way round.
- **J3**, the barrel jack, overhangs the right edge on purpose. Its pins are big and soak up heat: hold the iron on 4 to 5 seconds, until the solder flows into the hole.

# 31 Transistors: how and where



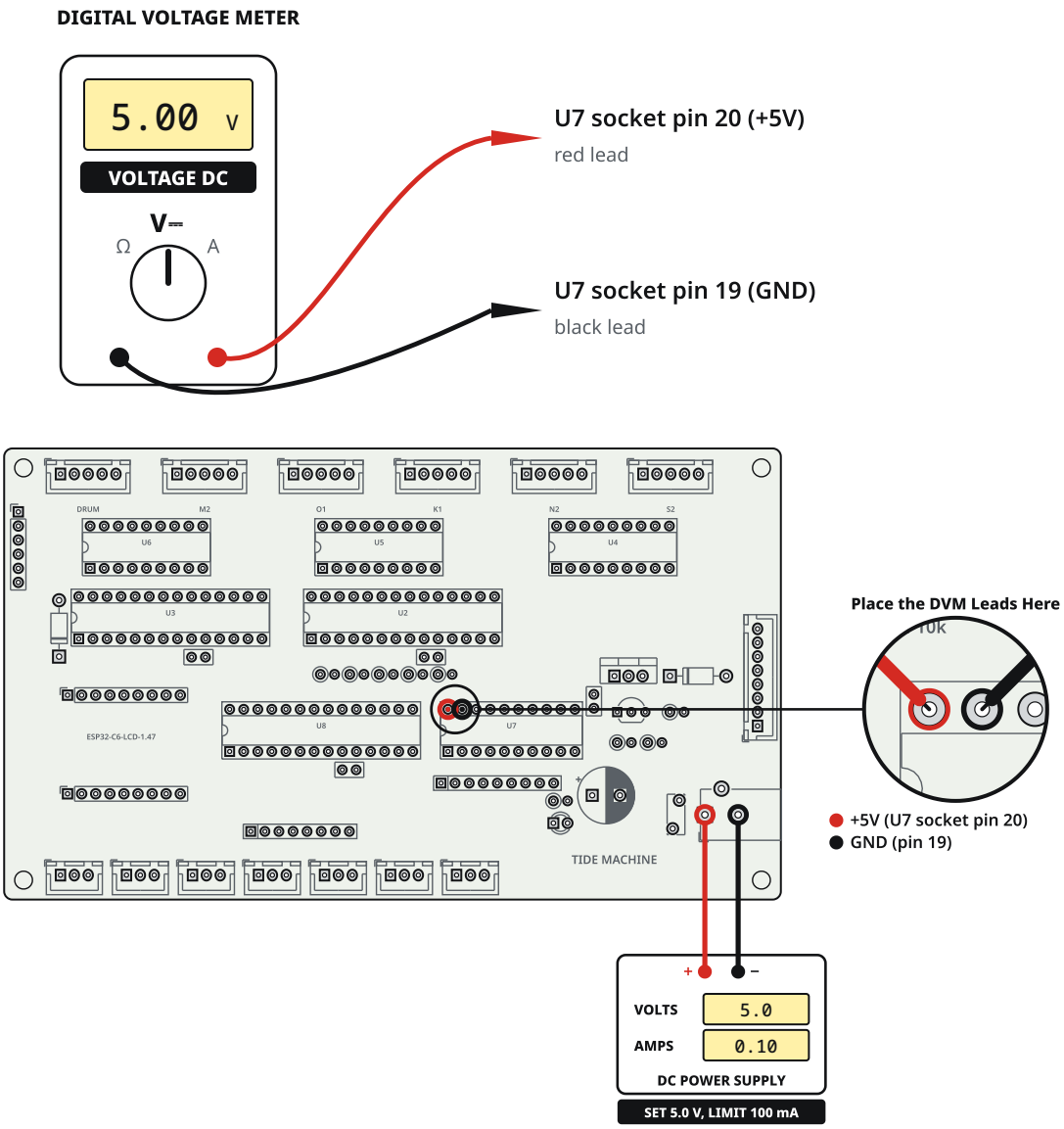




- **Q2** (2N3904): the **flat face** over the flat edge of the outline. Read the print before it goes in.
- **Q1** (IRLIB9343): **metal tab** to the **bold bar**, standing upright, no heatsink.

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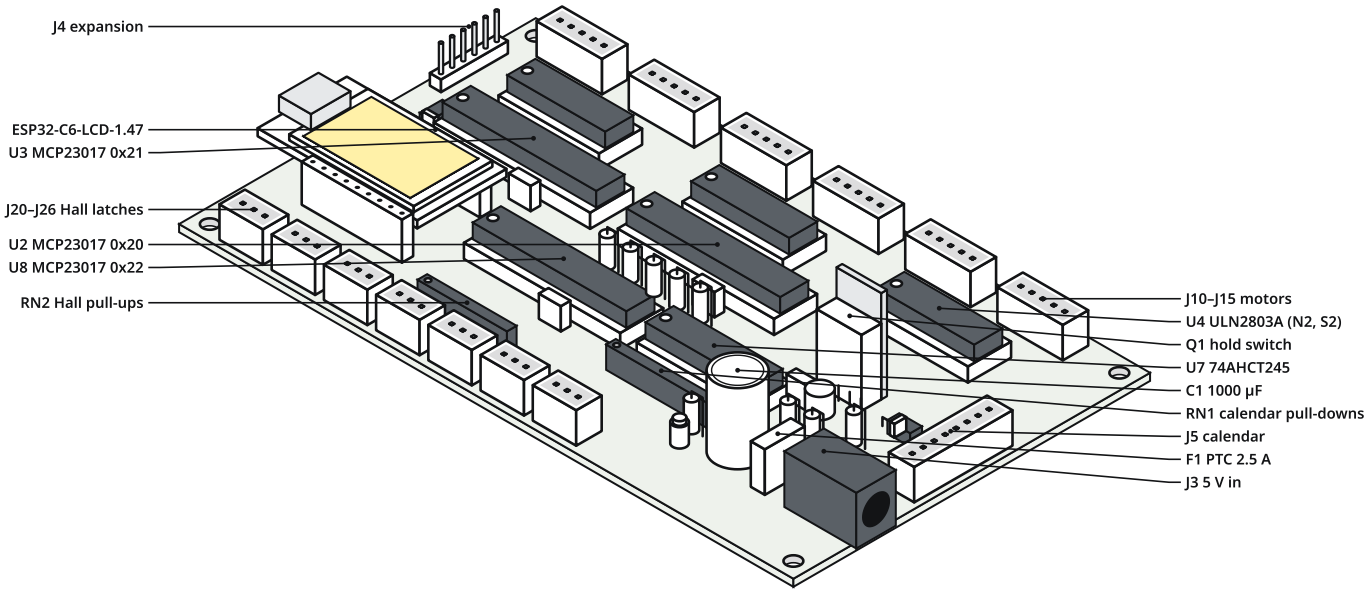
ⓘ Q1 switches the motors' 5 V (the HOLD rail) so the firmware can drop the holding current to 60% between steps. Fit it anyway; step 50 decides whether the hold is needed at all.

# 32 Quick check: first light




- **No chips, no C6.** 5.0 V into J3 through a **100 mA current limit**, centre pin +.
- **LED1 lights green.** The meter reads **4.95 to 5.00 V** in the empty U7 socket. Nothing gets warm.
- Power off. All good? On to section C. Not good? Section C's tests, one at a time, show which part is at fault.

# 33 Before you test

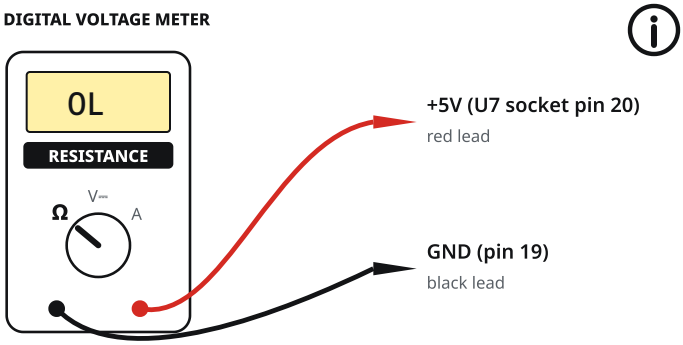



- Every box on every map ticked. The sockets stay empty.
- Turn the board over under a bright light. Every joint a shiny cone, no bridges, every lead trimmed.
- Scrub off the flux with isopropyl alcohol and an old toothbrush, and let it dry.
- Now Test 0. No chip and no C6 go in until section C says so.

# C Smoke test

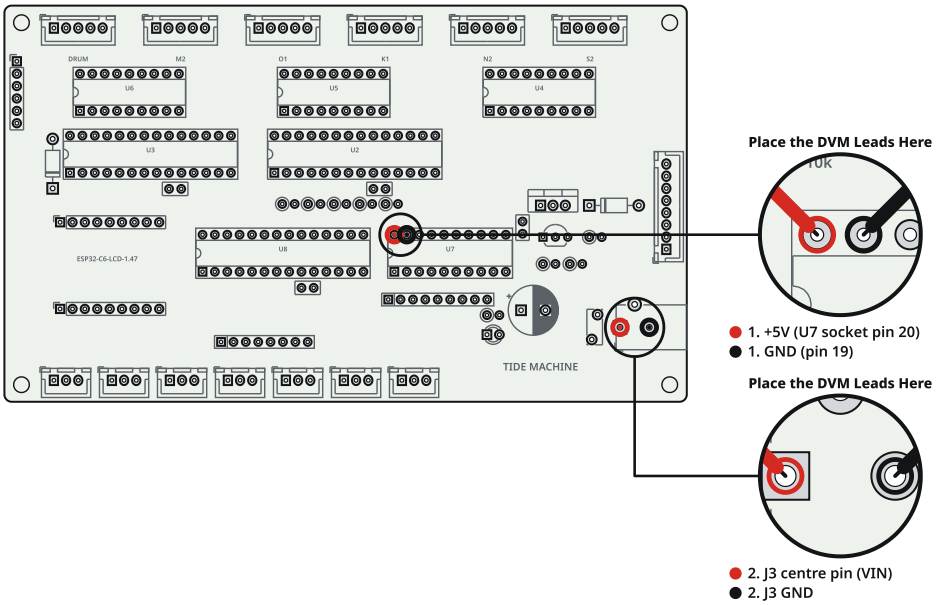
Nothing expensive goes in until every rail reads right. Use a bench supply at 5.0 V with a 100 mA limit, or a 5 V supply through a USB power meter.

## 34 Test 0: DON'T HOOK UP POWER YET – Look, then measure resistance



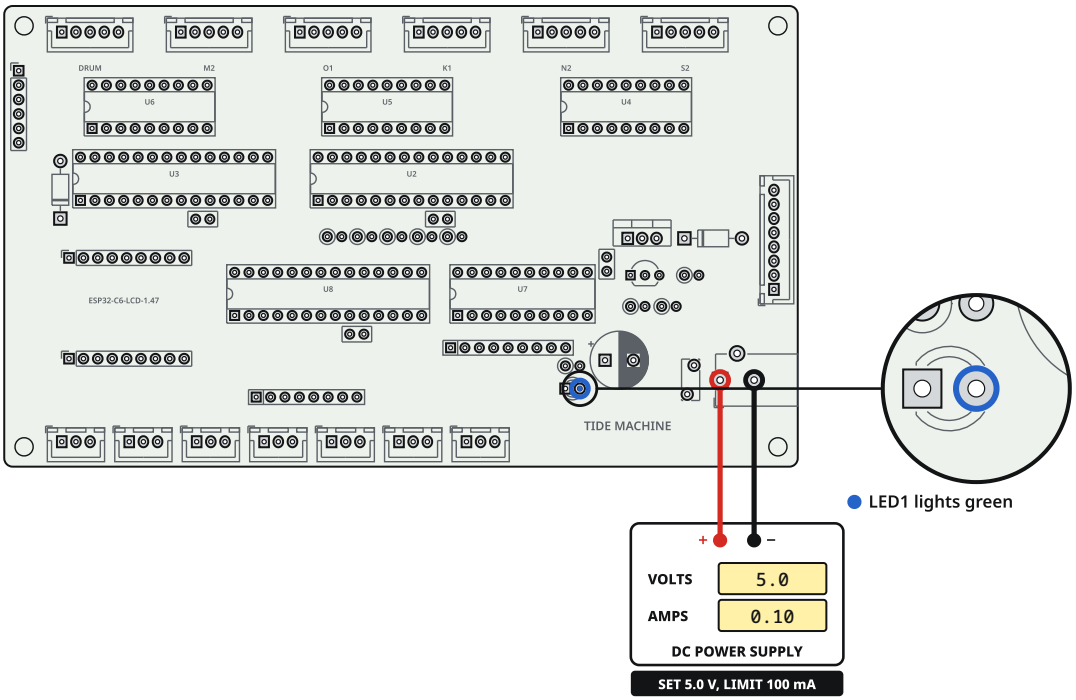

**How this works with no power.** On Ω the meter sends a small current of its own and reads the voltage it makes. Across the 5 V rail that current fills C1, so the reading starts low and climbs. A short never fills and stays low. The test: **does it climb, or stay put?**

Nothing in this board's 5 V input blocks one direction, so either lead works, but red on + is the habit to keep.




- Look at every joint under a lens: no bridges, no dull balls. Socket notches, diode bands, C1's stripe, Q1's tab: all as the silkscreen.
- **1.** +5V to GND **climbs** as C1 charges and ends above 1 kΩ. **2.** The same from J3's centre pin to GND. A reading that **stays** near 0 Ω is a short: **find it before any power goes in**.

# 35 Test 1: First power, current-limited

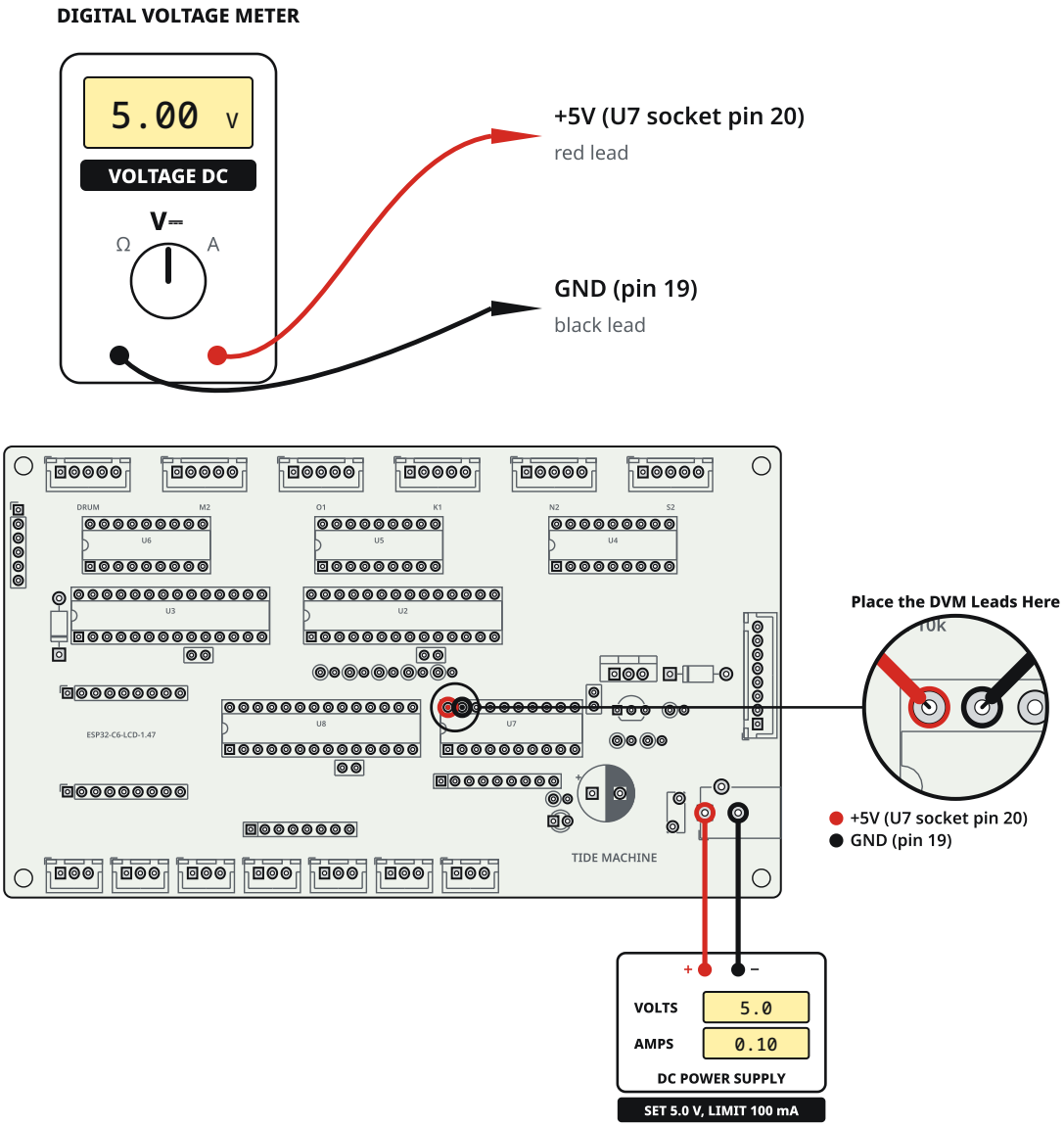



- At this point all the electronic components are installed. **EXCEPT DO NOT YET INSERT THE C6 OR THE CHIPS:** the C6 goes in at step 39, the chips at step 40, each only after the tests before it pass.
- 5.0 V into J3 through the 100 mA limit. **LED1 lights** and the supply's current reading shows about **3 mA**: (5.0 − 2.1 V) through R8's 1 kΩ.

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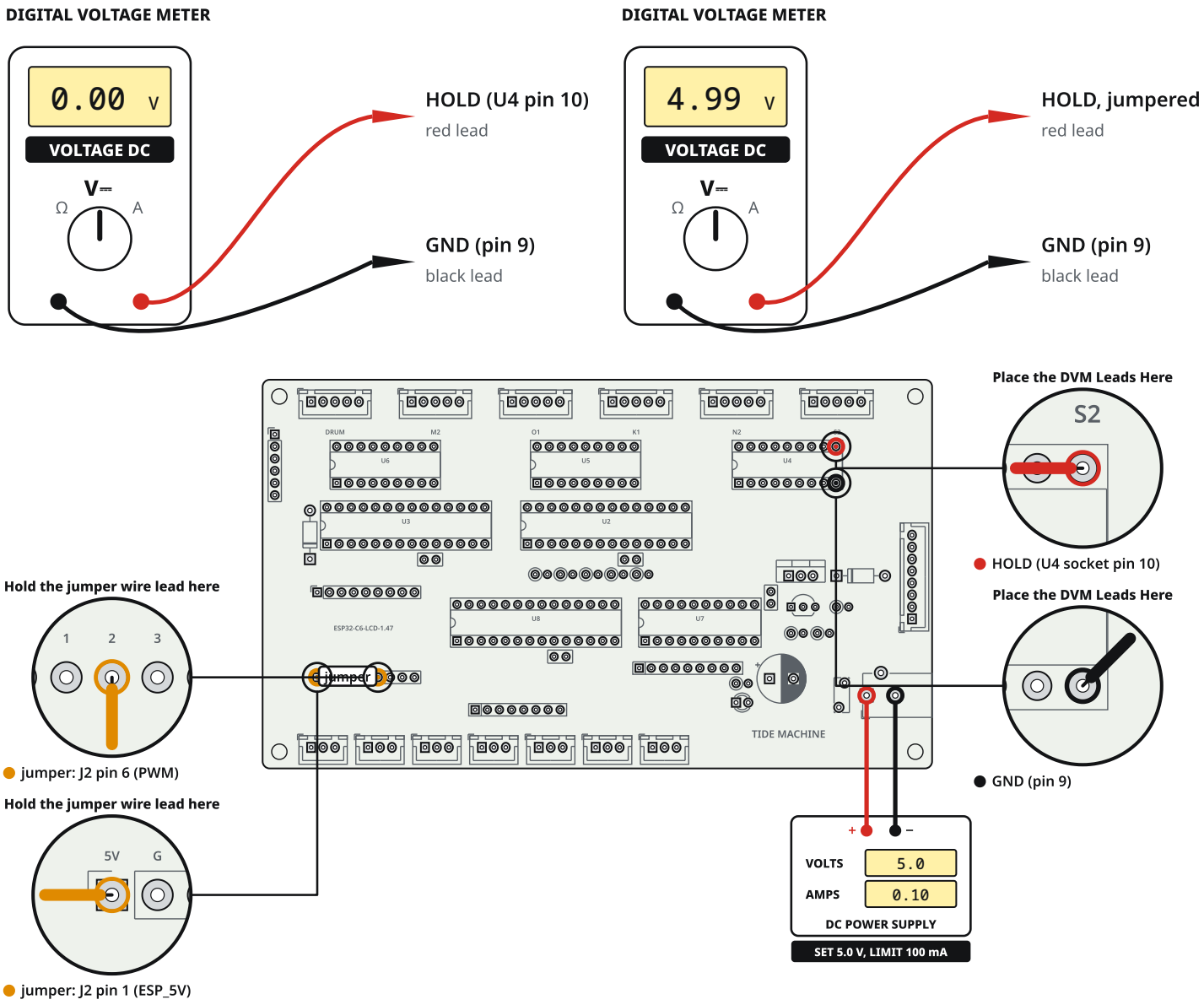
⚠ If the supply hits its limit, switch off at once and go back to Test 0.

# 36 Test 2: Verify 5 V reaches the board



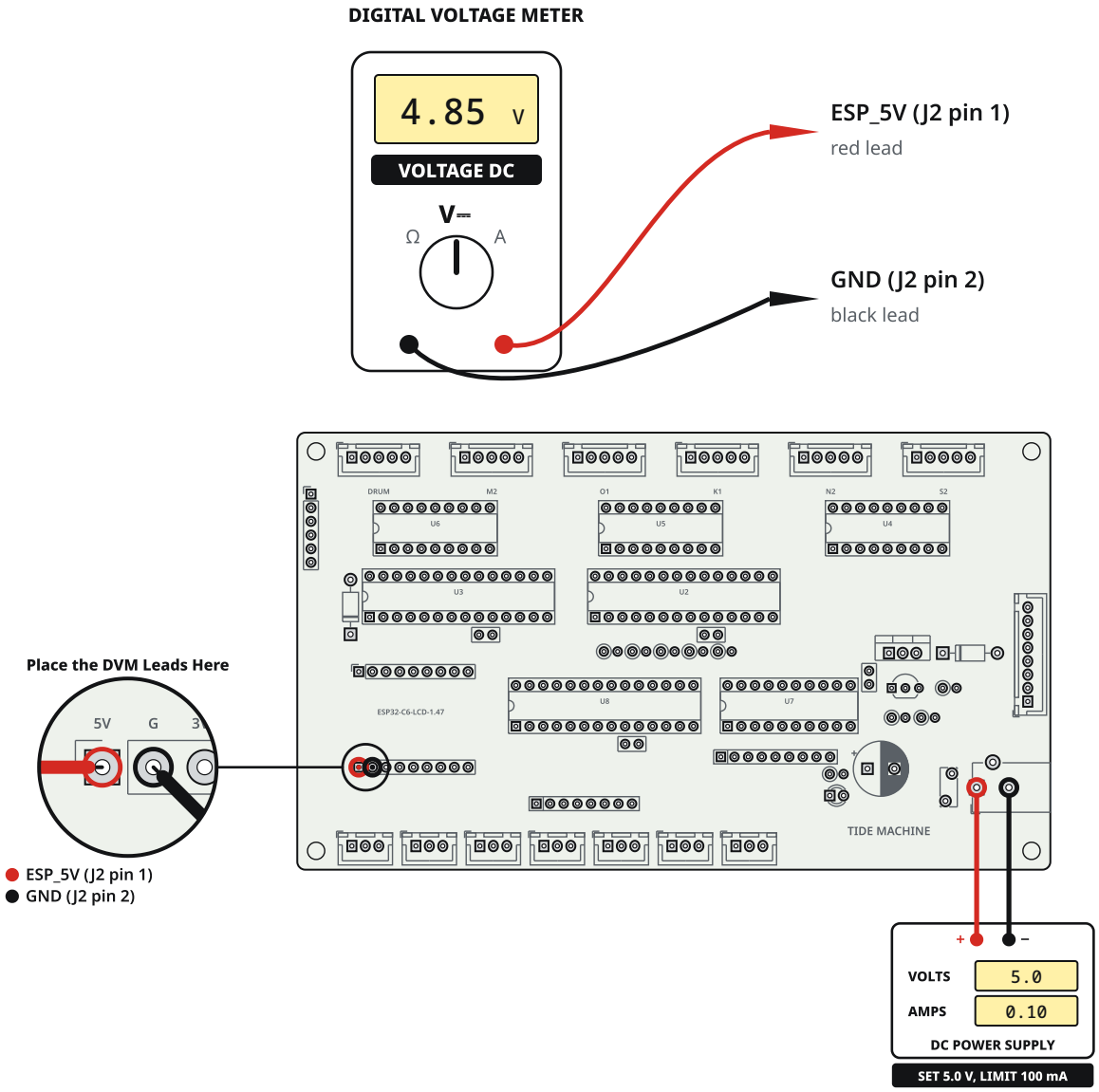

- **4.95 to 5.00 V**: the supply less F1's drop at 3 mA, which is next to nothing.

# 37 Test 3: The hold switch




- Power on, nothing plugged in: **HOLD reads 0 V**. R7 holds Q2 off, so R5 holds Q1 off.
- Jumper **J2 pin 6 (PWM) to J2 pin 1 (ESP_5V)**: HOLD rises to **about 4.99 V**. Remove the jumper.

# 38 Test 4: Verify the C6 gets its 5 V



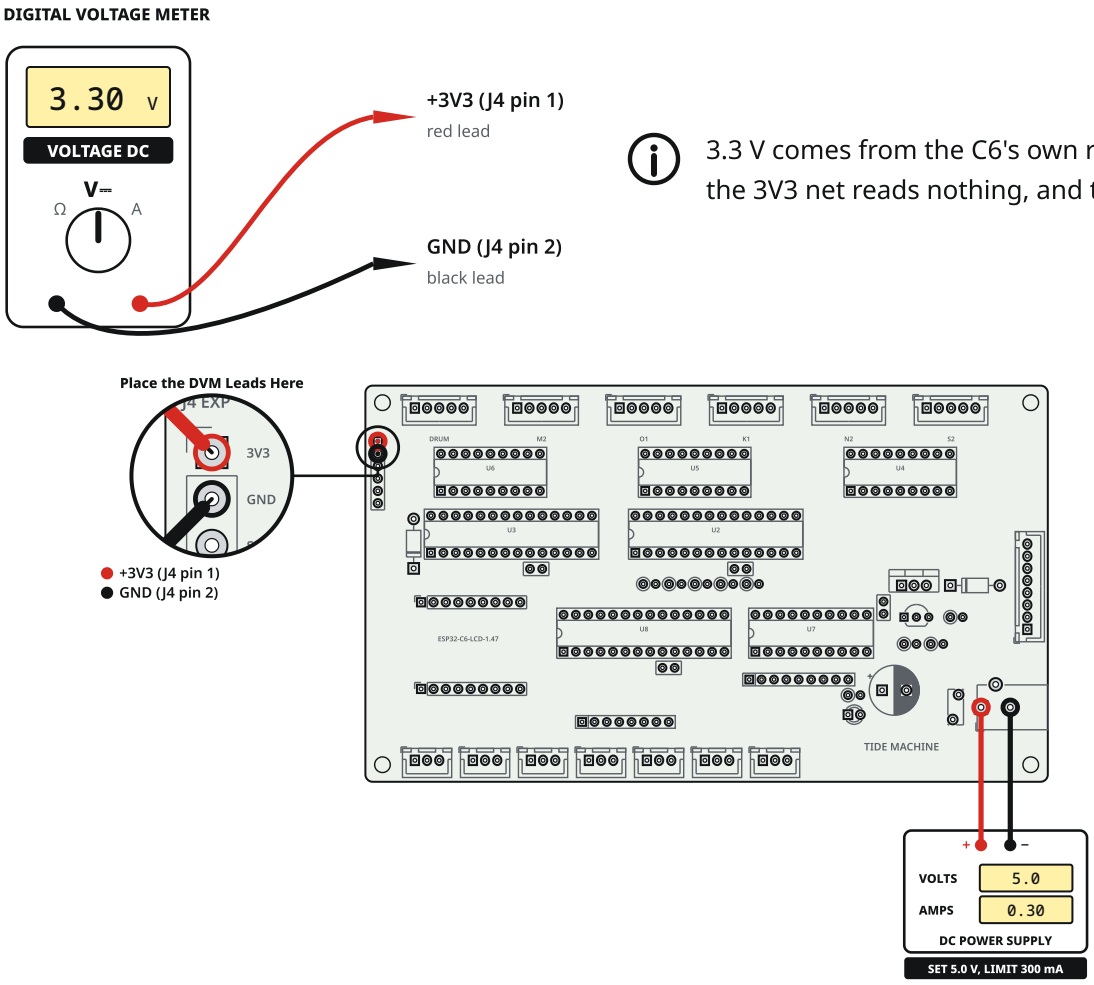

- **4.8 to 4.95 V**: +5V less D1's Schottky drop at almost no load.

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D1 keeps the supply and the C6's own USB apart, so flashing over USB with the board powered is safe.

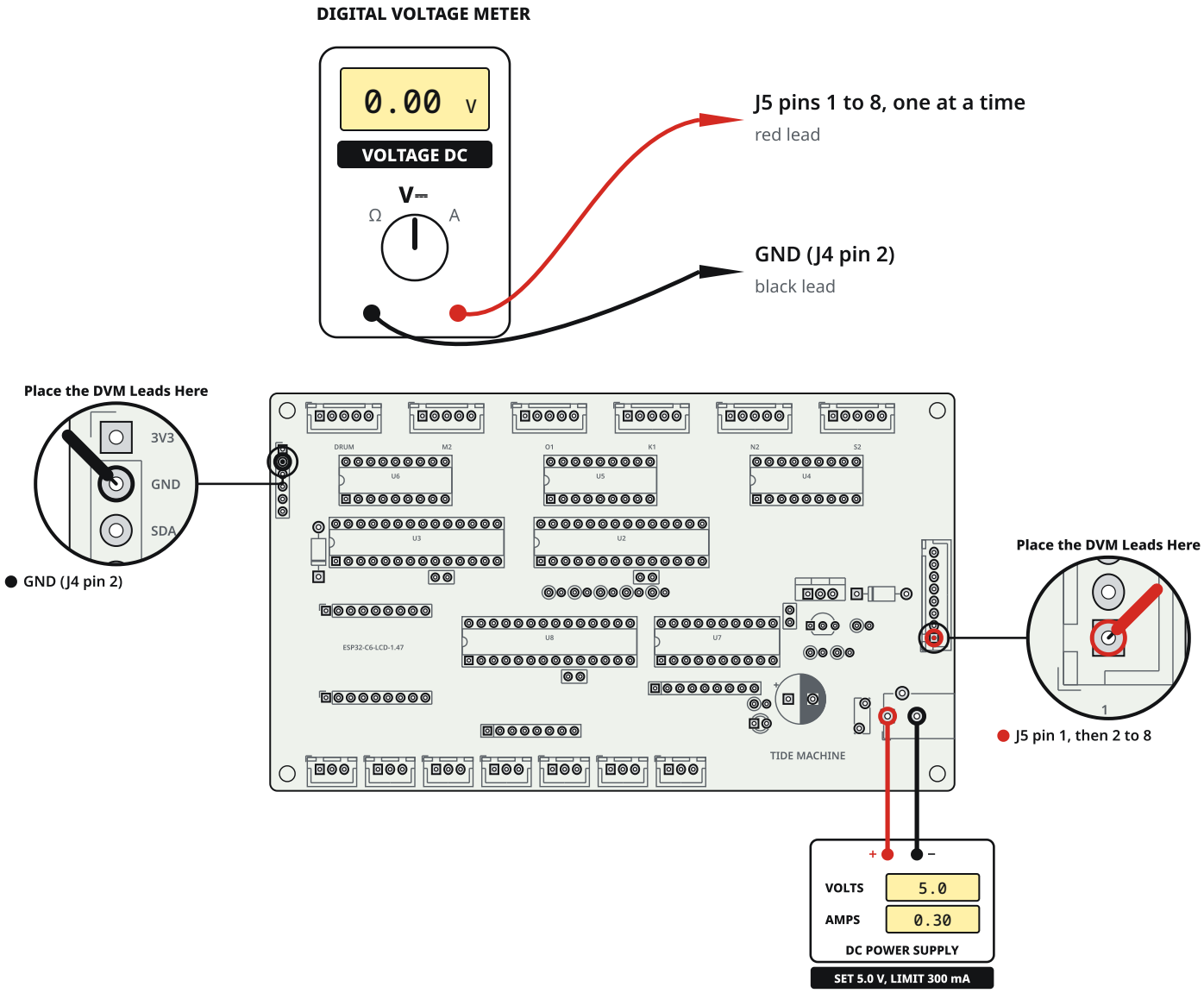
# 39 Test 5: Plug in the C6

fitting, then a check




3.3 V comes from the C6's own regulator. With no C6 fitted, the 3V3 net reads nothing, and that is correct.

- **Power off.** The C6 into J1/J2, LCD up, USB-C at the board's left edge.
- Raise the supply's limit to **300 mA**: the C6 and its screen draw more than the bare board.
- Power on. **3.30 V** on J4 pin 1. Also 3.3 V on SDA and SCL (J4 pins 3, 4), on INT_A and INT_B (J1 pins 6, 5), and on pin 2 of every Hall socket J20–J26.




- **Power off.** Chips into the sockets from step 26, notch to notch: **U2, U3, U8** MCP23017, **U4, U5, U6** ULN2803A, **U7** 74AHCT245. Check no leg folded under.
- Power on, limit still 300 mA. Every calendar line on **J5 reads 0 V** (RN1 holds U7's inputs low until the 0x22 expander is set up), and HOLD still reads 0.
- The current is the C6's own draw plus about 3 mA. Above 250 mA: switch off and look for a chip in backwards.

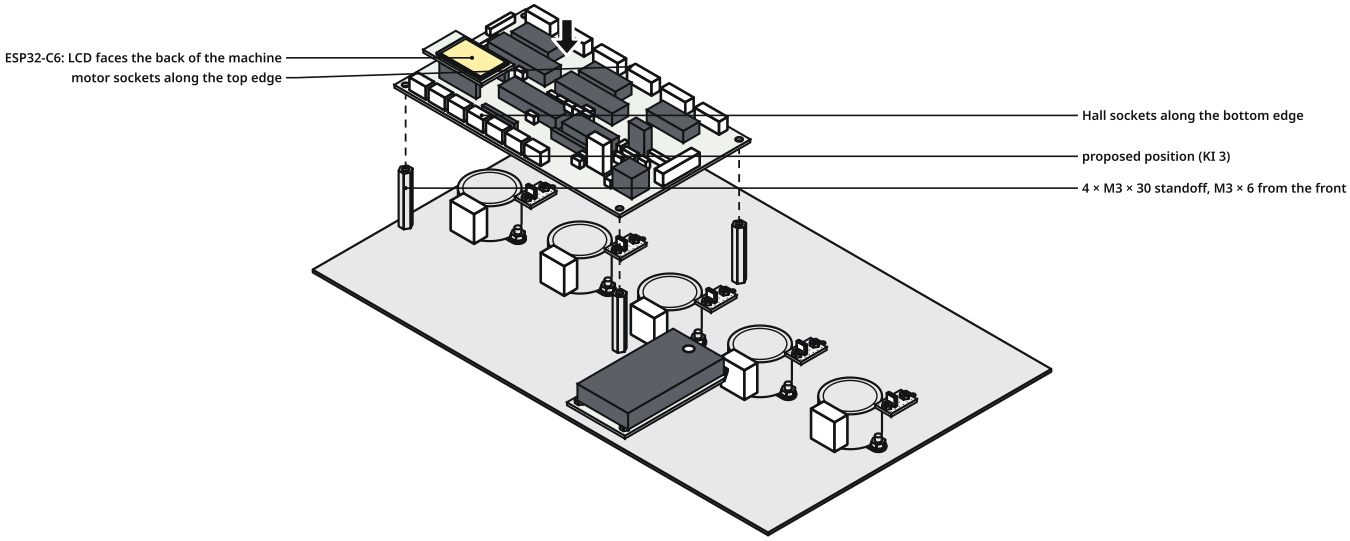
The I²C scan (0x20, 0x21, 0x22) is step 49: it needs firmware.

# D Wiring

The board goes on the back of the plate, 30 mm off it, clear of the motors. Everything plugs into its edges.

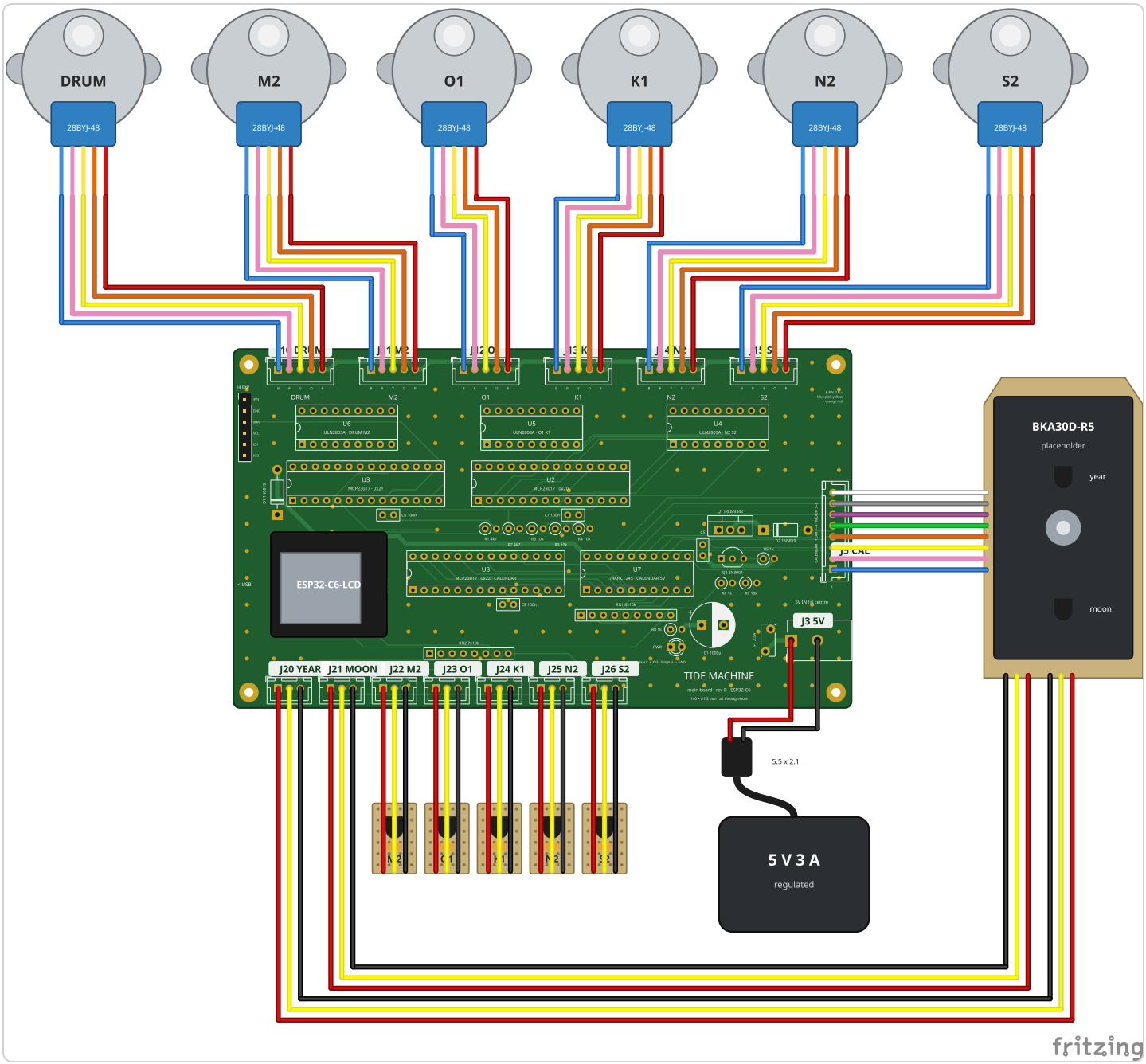
## 41 Mount the board on its standoffs

KI 3




- Four `M3 × 30` standoffs, `M3 × 6` from the front of the plate. The board on top, component side out, **silkscreen reading upright from behind**, four more `M3 × 6`.
- The board is 30 mm off the plate, the motors 20.5 deep: 8 mm under the lead ends.

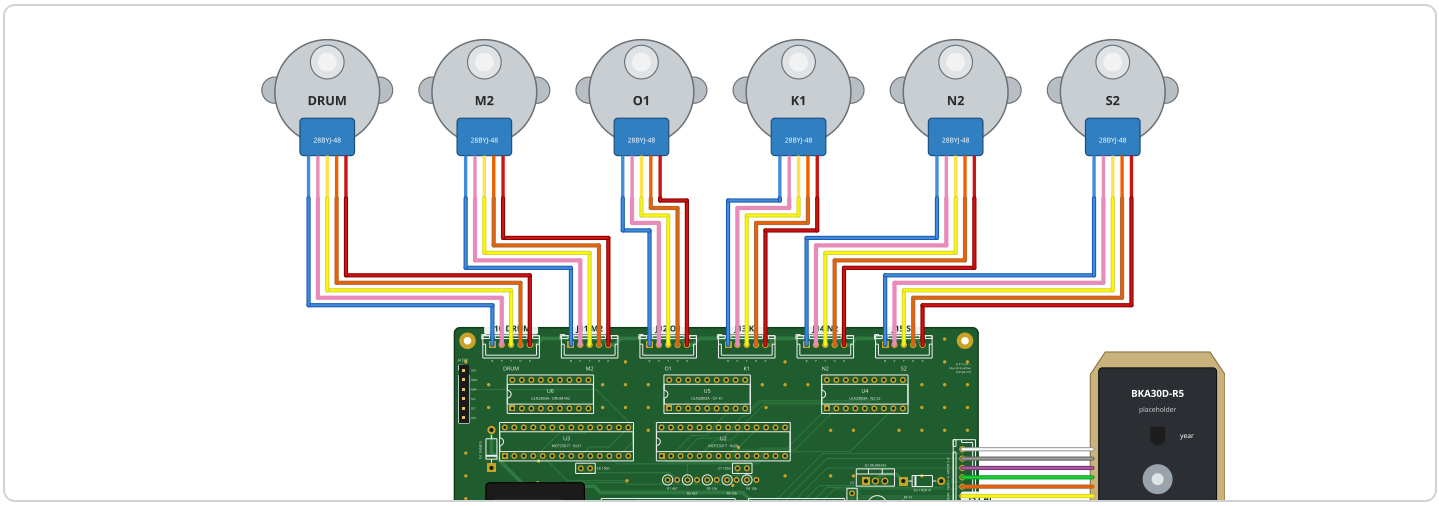
# 42 The whole loom

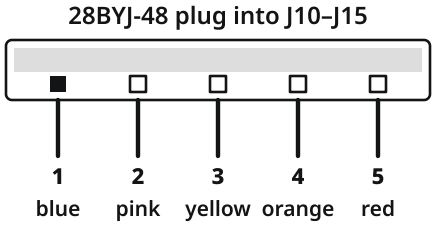



Fritzing, breadboard view. Source: `docs/assembly_guide/tide_wiring.fzz`, generated by `gen_fritzing.py`; the board is drawn from `pcb/tide_main.kicad_pcb`.

# 43 Plug in the motors

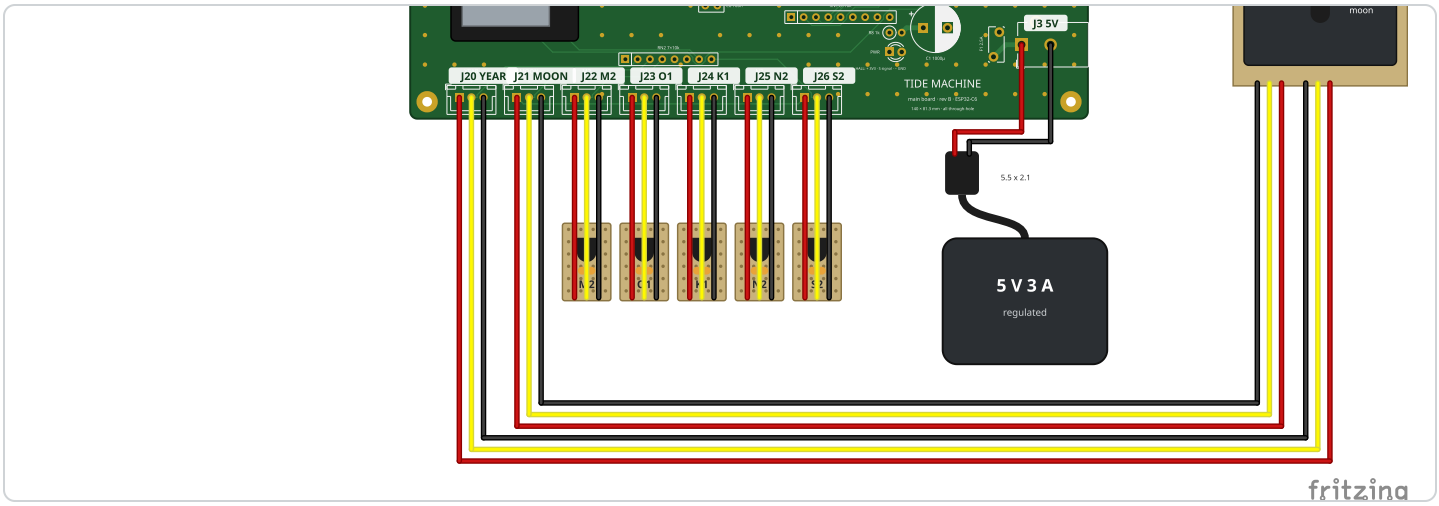
KI 8



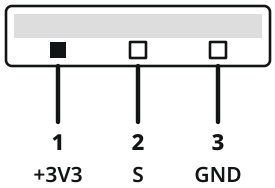




- Each 28BYJ-48's own plug into its socket: **J10 drum, J11 M2, J12 O1, J13 K1, J14 N2, J15 S2**, as the silkscreen says.
- The plug is keyed. Red is pin 5, the switched HOLD rail.

# 44 Crimp and plug the Hall leads




XHP-3 into J20–J26



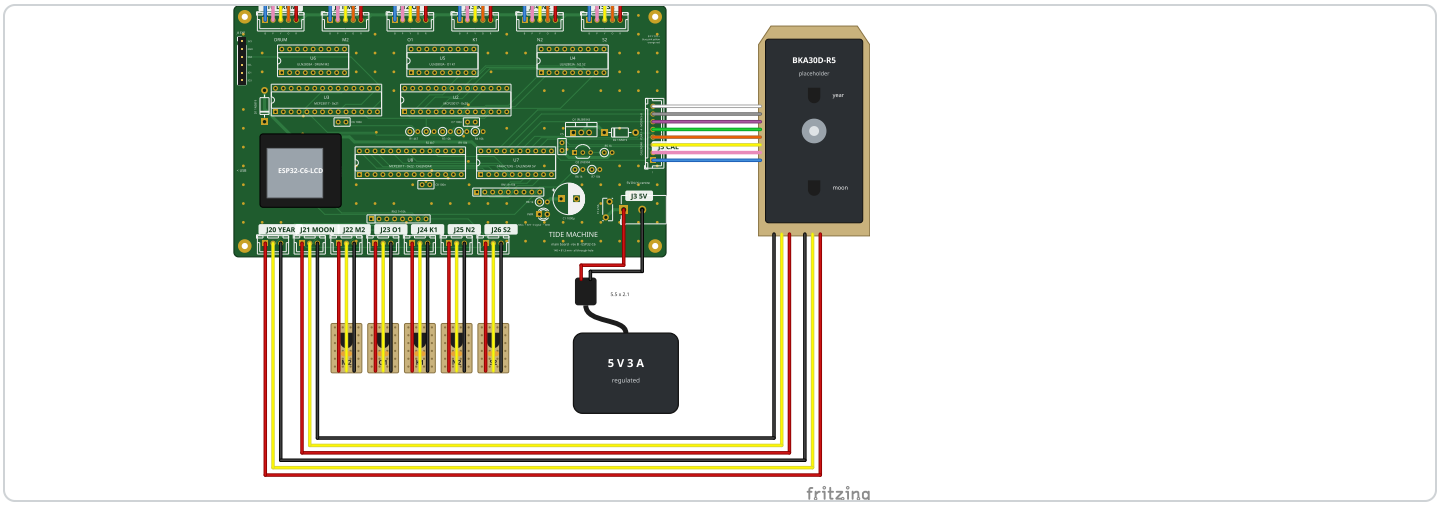

- Each 3-core lead into an `XHP-3`: pin 1 **+3V3**, pin 2 **S** (the latch output), pin 3 **GND**.
- **J20 year, J21 moon, J22 M2, J23 O1, J24 K1, J25 N2, J26 S2.**

---

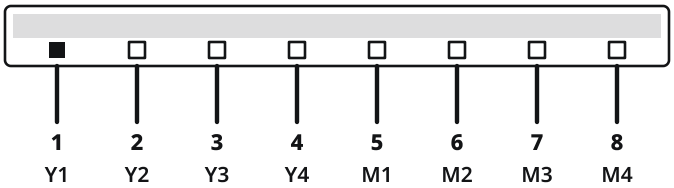
The pull-ups are on the board (RN2, 10 kΩ to 3.3 V), so each carrier needs only its latch and a 100 nF.

# 45 The calendar lead

KI 5

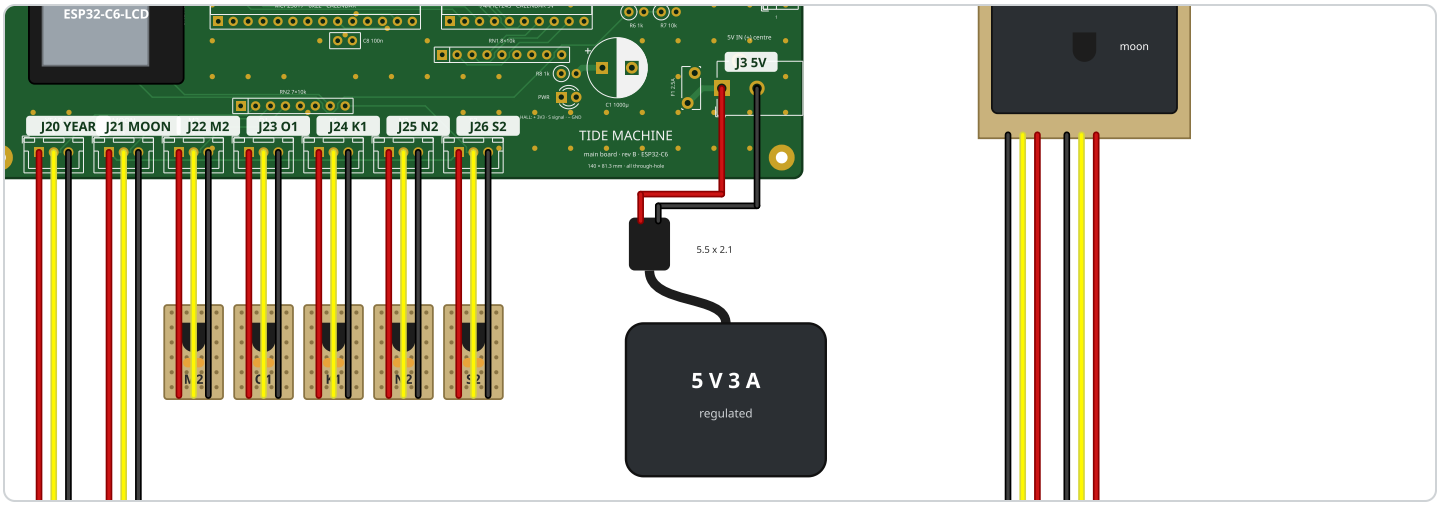



**XHP-8 into J5 (year coils 1–4, moon 5–8)**



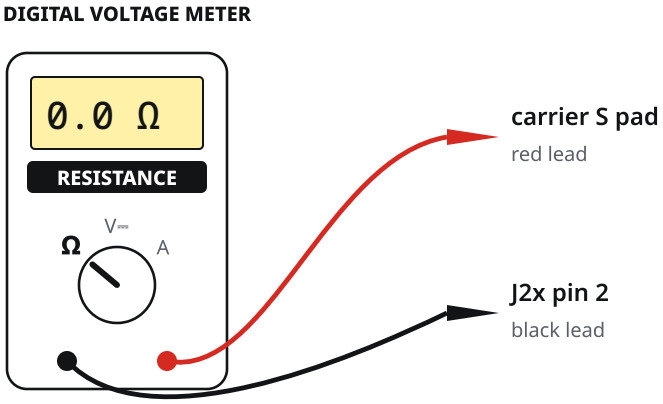

- 8-core from the BKA30D-R5 into an `XHP-8` on J5: pins 1–4 the year (outer) motor, pins 5–8 the moon (inner) motor.
- Match each motor's coil pins to Y1–Y4 and M1–M4 from its datasheet when it's in hand.

# 46 Power and USB




- A **5 V 3 A regulated** supply, 5.5 × 2.1 plug, centre positive, into J3. Peak draw is about 1.5 A with every motor energised.
- USB-C on the C6, at the board's left edge, for flashing and the serial console.

# 47 Pin check before power




- Power off. Beep every Hall lead end to end (+, S, −) and the calendar lead pin by pin.
- +3V3 to GND on each Hall socket must **not** beep.

# E First power-up and calibration

Each step is tagged **today** (works with what's in the repo now) or **planned** (needs firmware that isn't written yet).

## 48 Flash the firmware

planned

- Over the C6's USB-C. The firmware carries the tables from `pcb/design.py`: `MOTOR_BITS`, `HALL_BITS`, `CAL_BITS`, `CAL_HALL_BITS`, `ESP_GPIO`, plus a per-station inverse table from `kinematics.py`.

---

ⓘ The C6's LCD faces the back of the machine: planned as a status screen (time, phase of each station, last Hall edge).

# 49 I²C scan

planned

- Expect exactly **0x20, 0x21, 0x22** on SDA GPIO19 / SCL GPIO18. A missing address is a chip not seated or an address pin: U2 all low, U3 A0 high, U8 A1 high.

# 50 Holding-torque test

today

- One spare 28BYJ-48 on the bench, a 100 mm lever on its shaft, a known weight on the end: torque = m · g · 0.1.
- Unpowered first, then with one coil energised (red to 5 V, orange to GND).

---

ⓘ M2’s arm needs 14.5 mN·m. The datasheet’s 59–118 mN·m “friction torque” hints the gearbox holds that unpowered. If so, the PWM hold can go to zero between steps.

# 51 Check every motor’s direction

planned

- At 1000×, one station at a time: each turns the way its table says. Every shaft turns one way only, which is what keeps gearbox backlash a constant.

# 52 Hall edges

planned

- Each station’s latch reports one edge per turn through the 0x21 expander’s INTB (GPIO20); year and moon through 0x22 (GPIO23).
- Run each station past its edge ten times: the step count between edges must repeat exactly (2048 per turn at the output shaft, less the table’s shape).

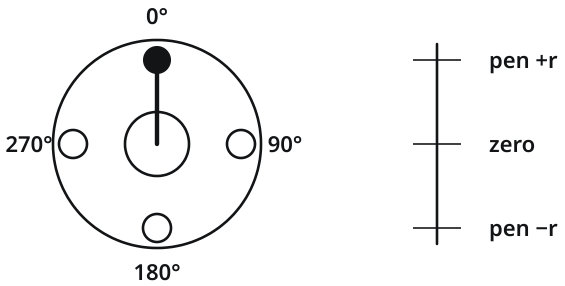
# 53 Home the calendar

planned

- Slowly (the gauge motor is weak): year dial to Jan 1, moon disc to new moon, each on its latch edge. Then to today’s date and the **true** moon phase.

# 54 Go-to-phase: check each arm’s radius

planned




- One station at a time: command its phase to 0°, 90°, 180°, 270°. The pen moves **±r** about its zero.

# 55 Set the pen zero

planned

- **Go-to-zero** (every output at 0), then turn the anchor screw on the pen bracket until the pen sits on the chart’s datum line.

---

The pen’s zero is set by line length and nothing else. Homing can’t crash the pen: its travel is bounded by Σr = 57.05 against 62 mm available.

# 56 Soak at 1000×

planned

- 90 minutes at 1000× (60 days of tide) against the Marigram Sandbox. No skips: every Hall edge arrives on its predicted step.

# F How to use

Set it running, load paper, read the dials.

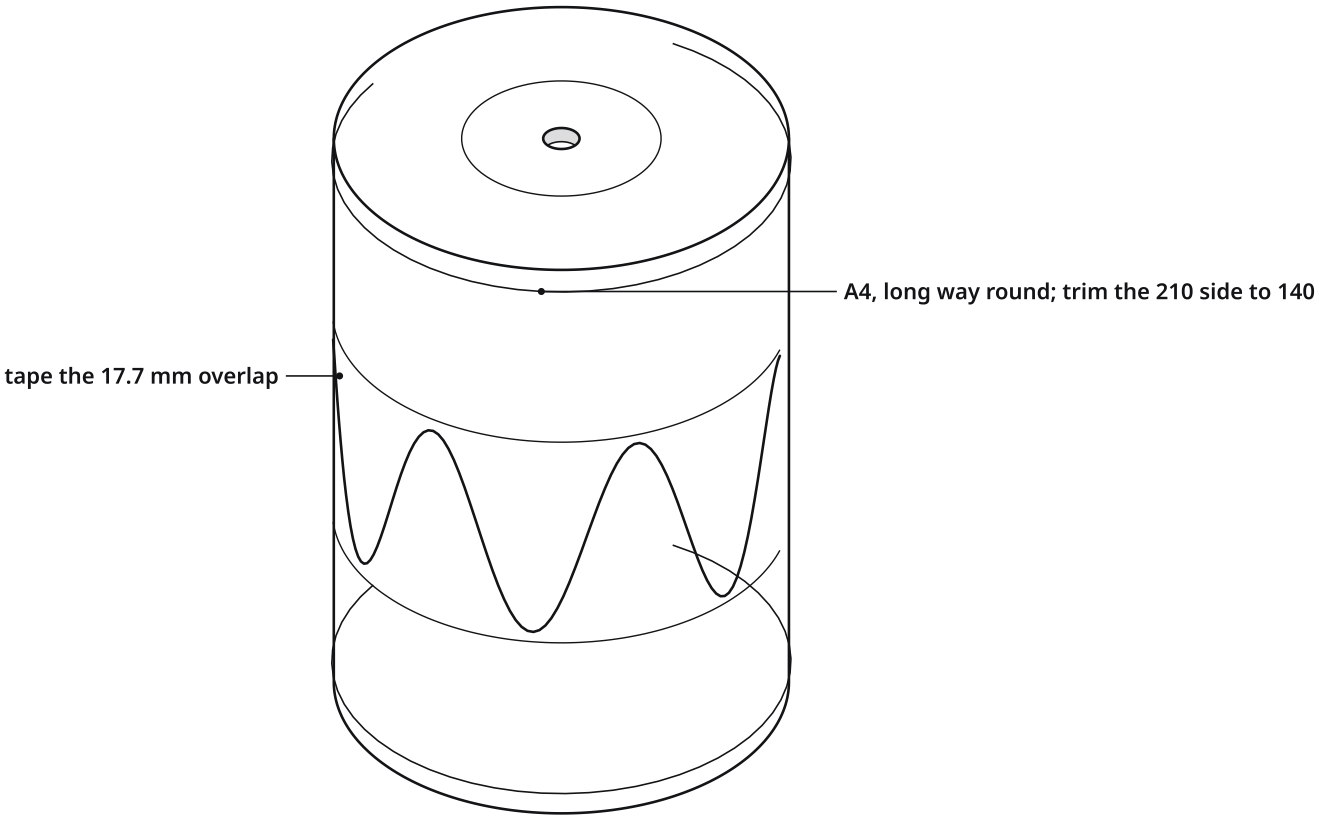
## 57 Start it

planned

- Power on: Wi-Fi, NTP, re-home every station, then run in real time. NTP again daily.

# 58 Load the chart paper

today




- A4 wrapped the long way round the 279.3 mm drum; trim the 210 side to 140. Tape the 17.7 mm overlap.

---

ⓘ Chart speed is still open: 7 days a turn (20.6 mm per tide) or 3.5 days (41.3 mm, a classic marigram). It’s a firmware constant.

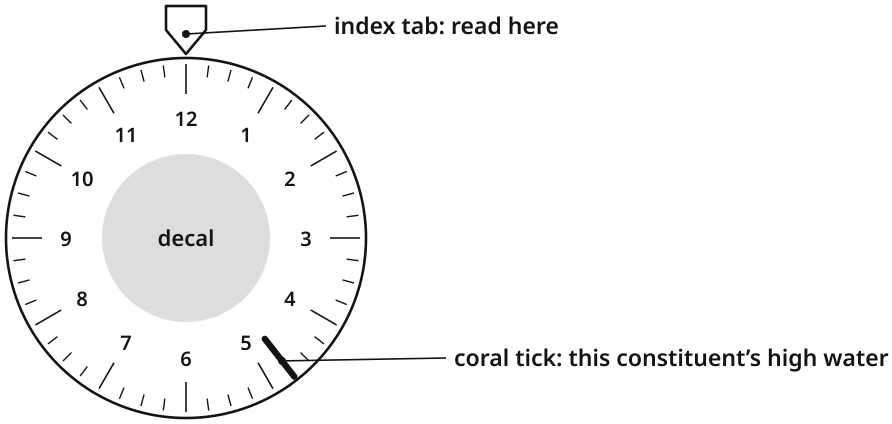
# 59 Replay at speed

planned

- The speed-up multiplier: 1000× replays 60 days in 86 minutes. Stay under 5000×, and re-home after any skip.

# 60 Read the dials

today

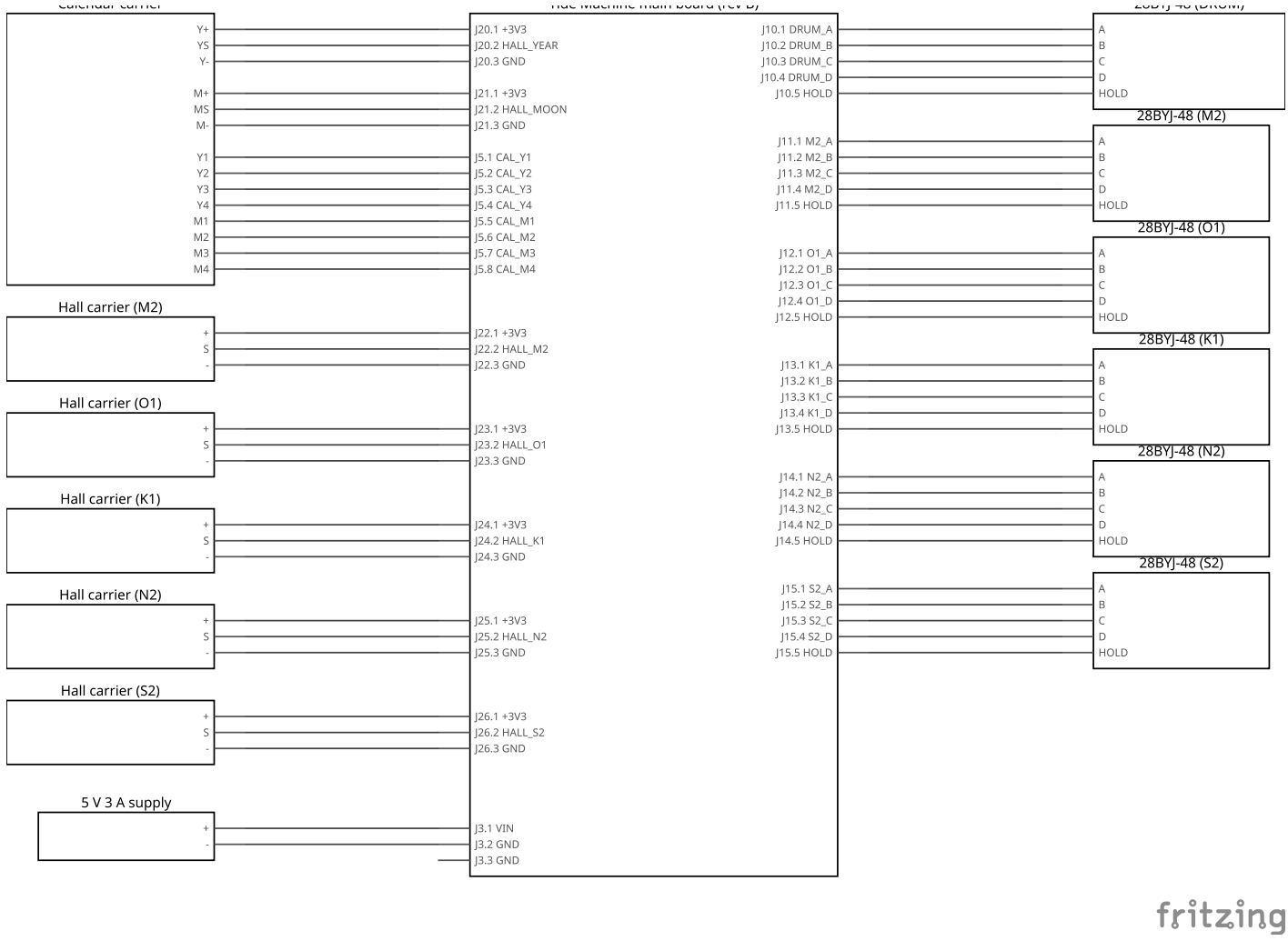



- Read each scale under its index tab; readings increase clockwise. **S2**: local mean solar time. **M2**: lunar hours. **K1**: sidereal time. **N2**, **O1**: hours of their own periods.
- The coral tick is each constituent’s high water. On M2 the arm itself is the pointer.
- Calendar: the hand reads the month on the year dial and days since new moon on the moon disc.

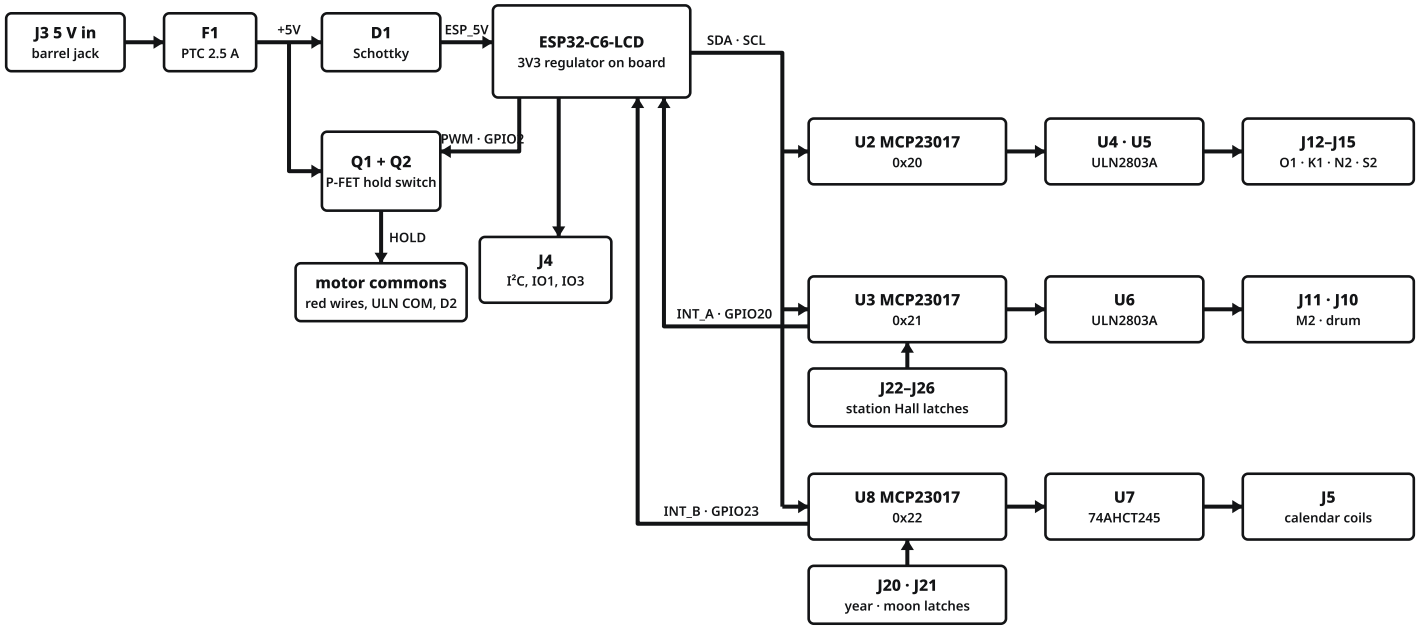
# G Appendix: schematics

The whole system, the board as blocks, then the board's full schematic.

## G1 · System wiring, schematic view



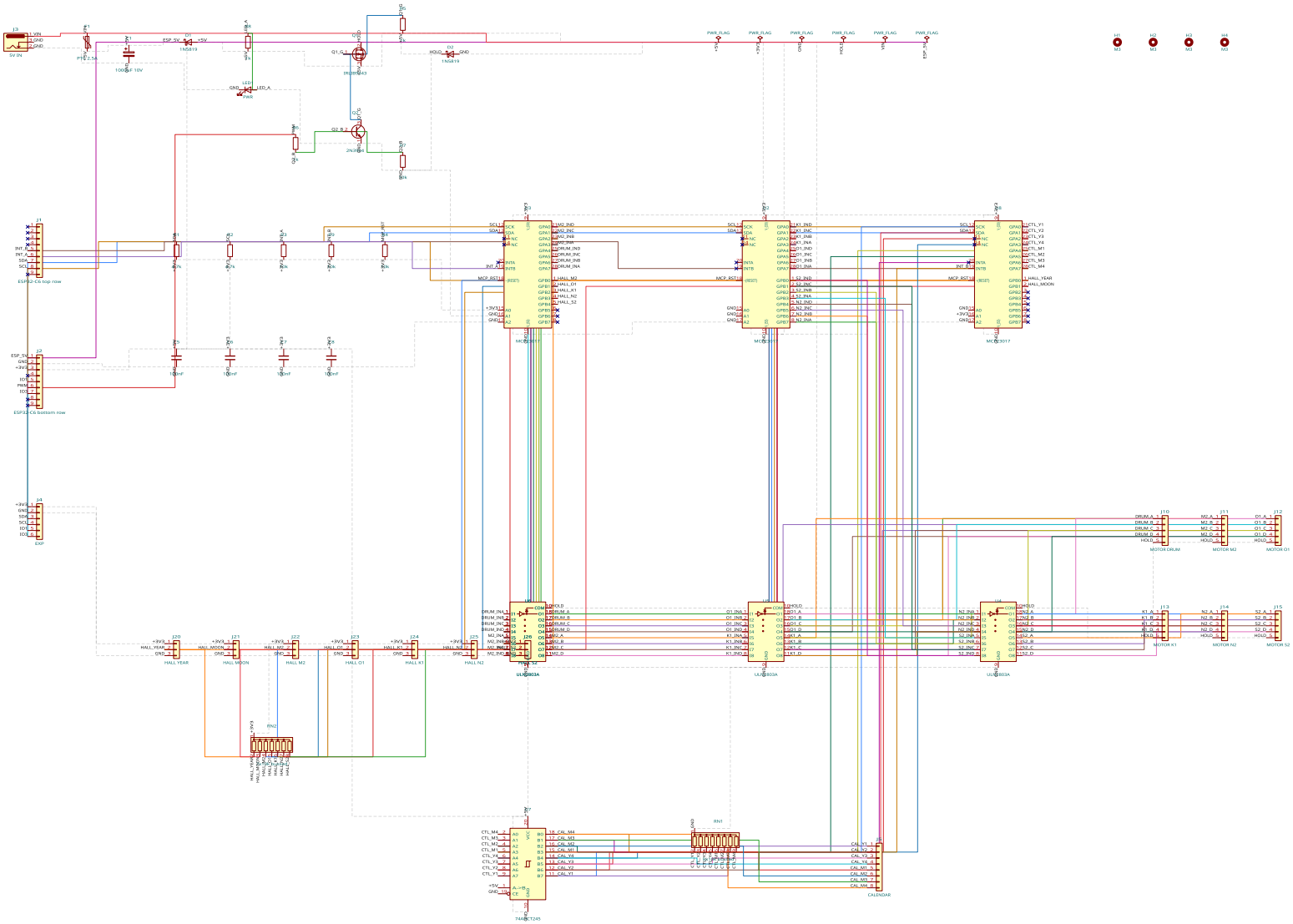
## G2 · The main board, logically




Signal names, not GPIO numbers, are on the silkscreen. Hall inputs have 10 kΩ pull-ups (RN2); the calendar lines have pull-downs (RN1).

Every block and net from `pcb/design.py`, the same data the board is built from.

# G3 · Main board, full schematic



The drawing is vector: zoom in as far as you like. Parts are joined by net label in KiCad; the coloured lines redraw those connections for this page (one colour per signal net, faint dashed grey for the many-pin rails). Generated from `pcb/design.py` by `pcb/gen_schematic.py`; ERC clean.